

ESQ Conference on Optical Telescopes of the Future This conference took place in Geneva between 12 and 15 December 1977. The time seemed ripe for a conference on this subject, for many ideas are in the air and certain projects in the United States which deviate markedly from the conventional telescope are already completed or in active study. The conference opened with a review of the astronomical case for large telescopes, an overview of the technological possibilities and the possibilities from space. A session followed on conventional large telescopes in which technical aspects of a number of existing large telescopes and the possible extension of conventional telescopes to larger sizes were presented. The following day was devoted entirely to incoherent arrays and multi-mirror telescopes. In the sense that most effort deviating from the conventional large telescope has so far gone in this direction, the neutral observer had the feeling that this represented the center of gravity of the conference. A wide variety of interesting solutions were presented with a collecting area up to the equivalent of a 25 m telescope and arrays with up to 100 telescopes. A session on special techniques fitting into no clear group was followed by sessions on coherent arrays and interferometers. This gave a broad review of current techniques and future possibilities and a comparison of optical and radio techniques. The last morning of the conference, concerned with image processing and live optics, showed clearly the tremendous gain to be obtained in overcoming the effects of seeing even without increase in instrumental size. A clear distinction emerged at this conference between the terms "active optics" and "active structure." The latter implies, for example, the active control of tilt or position of several mirrors to combine images; the former the control of the form of, say, a thin mirror. The importance of both possibilities became increasingly clear throughout the conference. The final session included a review of trends in detector developments. The rest of the session was devoted to a review of the astronomical implications of the contributions and discussions, followed by a panel discussion. The latter developed into a most lively general discussion with numerous participants representing very many viewpoints. While the consensus viewpoint seemed to support the view that the emphasis for post-conventional telescopes should lie in the incoherent addition of more photons from bigger systems, a strongly vocal minority was

clearly convinced that techniques using phase information should not be neglected. The discussion also inevitably brought up the vexed question of how an astronomer should or would like to work with future instruments. Thus ended a conference which seemed to have largely fulfilled its purpose: to encourage the debate on how instrumental funds in the future should be spent to best effect. The organizers thank all participants for making it such a stimulating event. Our thanks are due particularly to all the speakers who have enabled us to produce a virtually complete volume of proceedings within two months of the conference. Forthcoming ESQ Workshops Two ESO workshops have been planned during 1978 on the subjects of astronomical photography and infrared astronomy. As in the case of earlier ESO workshops, attendance is limited and by invitation only. "Modern Techniques in Astronomical Photography" This workshop will take place in Geneva on the CERN premises during May 16-18, 1978. About 50 participants are expected, mostly from European countries, but also from North and South America and Asia. The two principal subjects to be discussed are sensitization and calibration of photographic plates. There will also be a discussion about the copying of plates and use of colour photography in astronomy. Several participants will talk about the photographic work at their observatories and a number of new techniques will be reported. The proceedings will appear shortly after the conference. Further information may be obtained from R.M. West, ESO, c/o CERN, CH-1211 Geneva 23, Switzerland. "Infrared Astronomy" The ESO workshop on infrared astronomy will be held on the island of Uto on June 20-22, 1978. About 40 European astronomers active in infrared astronomy will be invited in order to discuss the scientific framework, the research planning and the instrumental development which ESO should foster in this area. The programme of the workshop includes: Review talks on astrophysical problems where infrared observations are of particular value, Review talks on the present status of various instrumental techniques in the field of IR spectroscopy and photometry, -Reports on IA space project and on other European plans. More detailed information may be obtained from P. Salinari, ESO, c/o CERN, CH-1211 Geneva 23, Switzerland. The Sculptor Dwarf Spheroidal Galaxy The first visiting astronomer to use the 3.6 m telescope in October 1977 was Dr. Steven van Aagt from the Astronomical

Institute of the Nijmegen University, the Netherlands. At that time the object for his study, the Sculptor dwarf galaxy, passed close to the zenith of La Silla at midnight. He obtained prime-focus photographic plates for the study of variable stars in this nearby galaxy. A very large reduction work is connected with this kind of astronomical research, and it is therefore not yet possible to give detailed results, but Dr. van Agt here discusses the reasons for investigating the Sculptor dwarf galaxy. Forty years ago Harlow Shapley announced in the Harvard Bulletin No. 908 the exciting discovery of "A Stellar System of a New Type" in Sculptor. The new system showed up as an assembly of hazy images on an exposure with the 24-inch Bruce telescope of the Boyden station in South Africa. The first confirmation of the reality of the object came from a plate obtained by S. I. Bailey in 1908, on which a faint patch of light was seen at the position of the Sculptor system. Bailey obtained this plate during a site-testing expedition with a 1-inch telescope and a total exposure over five nights of 2316! Additional observations with the 60-inch telescope resolved the individual stars and ruled out the possibility that the Sculptor system could be an extended cluster of galaxies. Dwarf spheroidal galaxies are generally known by the name of the constellation in which they appear. Within approximately 250 kpc, seven Sculptor-type systems are now known including the recently-discovered dwarf in Carina. In addition, three dwarf spheroidal galaxies have been discovered close to the Andromeda nebula. Within the Local Group there are now ten Sculptor-type systems known. Since these objects are difficult to detect behind the stellar foreground of our Galaxy it is not likely that this number is free from selection effects. At the time of the first discovery the interest of astronomers was focused strongly on the significance of the shapes of galaxies. Nowadays the dwarf spheroidal galaxies, and especially the nearest, offer a unique possibility to study the evolution of isolated stellar agglomerates. The low surface density of the stars permits inspection of the individual stars, also in the center region of the systems. It gives a unique possibility to trace, in a complete survey, all the variables, of which there are many, through the whole system. The Sculptor dwarf spheroidal galaxy is located at a distance of 78 kpc (250,000 light-years). This is derived from a mean, apparent luminosity of 20.13 magnitude in B for the RR-Lyrae variable stars. On the sky the Sculptor system

has a considerable size. The more than 600 variables which have now been discovered cover an elliptical area with a major axis of about two degrees, corresponding to a linear dimension of 2.7 kpc. The positions and identifications of the variables are now in press. New photographic observations have been obtained by the author at the prime focus of the ESO 3.6 metre telescope at La Silla in October 1977. The plates reach beyond magnitude 21.5 in 40 minutes. The aim of the programme is the determination of the periods and the luminosities for a selection of the variables in the 16 arcminute field of the 3.6 metre telescope. The field on which the plates are exposed in this part of the programme contains a photoelectric sequence as well as a secondary photographic standard sequence. At present the plates are being reduced at the Department of Astronomy of the University at Nijmegen, where a semi-automatic iris-photometer and a unique projecting Blink comparator are available. Although many characteristics of the stars in dwarf spheroidal galaxies are very similar to those of the stars in globular clusters, there are also significant differences. One is the occurrence of bright cepheids which do not follow the Period-Luminosity relation of population II cepheids. In the Sculptor system and in other dwarf spheroidal galaxies these anomalous, so-called BL Her stars are brighter than the cepheids in galactic globular clusters by approximately 0.5 to 1.0 magnitude at the same period. Similar anomalous BL Her stars are likely to be present in the Small Magellanic Cloud. It has been suggested that these stars belong to a younger population than the majority of stars in the same dwarf galaxy. In this hypothesis the galaxy itself was formed independently after the collapse of our Galaxy. If higher masses are assumed for the anomalous BL Her stars, another hypothesis put forward to explain the existence of these stars is that mass-transfer is taking place within binary systems. The observational evidence, however, is not sufficient and in general the knowledge about the stellar content and more specifically about the numerous variable stars is still incomplete.

Quasars and BL Lac Objects as Active Nuclei of Giant Galaxies Are we beginning to understand the nature of the quasars and the equally mysterious BL Lacertae objects? Are they nothing but extraordinarily bright galaxy nuclei? The observed similarity and continuity between the active nuclei of Seyfert type I, broad-line radio galaxies and quasars began to be fully exploited only

around five years ago. The possibility of intrinsically similar physical processes in both types of objects was raised earlier, but the basic idea of quasars and BL Lac objects as very powerful nuclei of large galaxies is only a recent one and not yet entirely accepted. The similarity between the Seyfert type I galaxies and the quasars are (i) in their optical line spectra characterized by very broad allowed lines (typical velocities of variable on a time-scale of months, and narrower forbidden lines with a large range of ionization stages, and (ii) in their variability at optical frequencies on a time-scale of weeks. The radio sources are also found to be very variable. The BL Lac objects have a featureless optical continuum spectrum (yet in some cases very weak emission lines have been detected) and they are characterized by an extreme variability at radio frequencies, down to time scales of days. The current definition of quasars (which also applies to BL Lac objects) is that they have star-like images on direct plates. Yet nebulosities or "fuzz" associated with quasars were known since the beginning of quasar research, i.e. around TON 256 and 3C 48, and were the subject of discussion and puzzlement. In particular the large extent of 100 kpc accompanying 3C 48 illustrated in the Figure did not suggest a normal galaxy. A photographic programme was undertaken to attempt to detect underlying galaxies centered on quasars. The quasars are so bright that their light could swamp that of the underlying galaxy. Thus the latter could be detected only for quasars of small redshift and if its image size is greater than that of the quasar. These observations were consistent with the hypothesis of quasars as active nuclei in galaxies. Indeed those quasars which were predicted to show underlying galaxies did so and those which were predicted not to show underlying galaxies did not, with the exception of 3C 48. However, photometric studies of faint envelopes of galaxies, also in progress in the early 1970s confirmed the existence of large envelopes. Both elliptical and spiral galaxies were found to be surrounded by large envelopes,  $\sim 100$  kpc for elliptical galaxies and not significantly smaller (by a factor of 2) for spiral galaxies (when compared at same integrated luminosities). Further one must emphasize that less attention was devoted to Seyfert galaxies than to quasars before 1968. Few "extreme Seyfert galaxies" of redshift above 0.01 were studied then. Some extreme Seyfert-type galaxies at redshift close to 0.05 were observed

among the objects in the Zwicky lists. All these active nuclei were indeed surrounded by nebulosities. Yet for most of them, a galaxy of stars and cold interstellar gas was not brought into evidence. The next step necessary to definitively solve the problem of underlying galaxies required spectroscopic observations of these nebulosities. Recently, such observations have revealed two types of nebulosities: (i) those dominated by a strong emission-line spectrum, and no detection of an intrinsic continuum (typically 3C 48), (ii) those characterized by a spectral energy distribution and absorption lines consistent with that of a normal galaxy of stars (typically BL Lacertae). In all cases, the redshift for this nebosity is very close to that of the active nucleus. This appears to rule out gravitational redshifts for quasars. Spectroscopic observations of the nebosity around the quasar 3C 48 were reported in 1975. Other quasars, and also the Seyfert galaxy 3C 120 were studied and their nebulosities exhibit the same type of strong emission-line spectrum. The two more intense optical lines are [O II] 3727 and [O III] 4959, 5007; [O III] is stronger than [O II] and much stronger than H, with in some cases [O III]/H as high as 20. This type of spectrum is unusual for a galaxy. It cannot be accounted for by H II regions heated by main-sequence stars, whatever the abundances of heavy elements. Hard UV or collisional heating is required. The hard radiation, 50 eV, emerging from the active nucleus is a possible energy source for the nebosity. The observed line spectrum could then be achieved if the gas density is low, from 0.03 to 3 cm. At a cosmological distance the nebosity is then similar in dimensions and mass to the extended neutral H disks around spiral galaxies. Other possible models involve denser gas, thus very clumpy material, i.e. dense filaments, heated by the emerging UV radiation from the active nucleus. A strong controversy about spectroscopic observations of the nebosity associated with BL Lacertae took place in 1974-75. The intrinsic continuum spectra of such nebulosities are very difficult to observe, due to their weakness and to the strong contamination of nucleus light in the observed annular apertures. The detection of typical absorption features, such as Ca II K and Mg I is a crucial point. At least two such detections have been made: for BL Lacertae and for the quasar PHL 1070. In both cases the extended light surrounding the active nucleus is consistent with a luminous galaxy of stars. There is a large number of nebulosities for which

only a featureless continuum spectrum has been detected. The magnitude of the nebulosity is then consistent with that of a large galaxy. Quasars and BL Lac type objects can now be more firmly identified as active nuclei of giant galaxies. For BL Lac type objects and some quasars, the surrounding nebulosity is entirely consistent with a giant elliptical galaxy. For quasars such as 3C 48, or Seyfert galaxies such as 3C 120, the nature of the "surrounding galaxy" is not as clear. The emission from the ionized gas is much larger than would be that of the stars and only a very high sensitivity would allow the detection of the intrinsic continuum and of absorption lines. Another possible approach, possible with present-day techniques for extreme Seyfert galaxies, would be the determination of a rotation curve within the nebulosity from the brighter emission lines.

CHIRON: A New Planet in the Solar System

Last October, found a new planet in the solar system. Comparing two plates from the 48-inch Palomar Schmidt telescope in a blink microscope, he noticed a small trail of a moving object. From these plates and others which were obtained on the following nights, it soon became obvious that the new planet had an exceptionally slow motion. At opposition the motion of a planet is inversely proportional to the distance and a first estimate put it at about the distance of Uranus, almost 3,000 million kilometers away. When more observations became available, it was possible for me to confirm this distance and to establish the orbit. Extrapolating backwards, found on old plates in the Harvard plate library, obtained in 1895, 1941 and 1943. Some further observations from Palomar helped to improve the orbit, and it is now known that it is a unique object in the solar system. It moves in a rather elliptical orbit ( $e = 0.38$ ) with perihelion just inside the orbit of Saturn and aphelion close to that of Uranus. It was actually discovered a few years after it had passed through the aphelion and will become as bright as magnitude 14.5 in 1996 when it again reaches perihelion. The orbital period is just over 50 years. For the benefit of the eagle-eyed readers, we here show two plates of it obtained with the ESO Schmidt telescope on January 9 and 10. The plates were exposed during 30 minutes rather low in the western sky, just after sunset. At that time the planet was nearly stationary, near its smallest right ascension. The seeing was bad, probably around 4-5 arcseconds on both occasions and the images are therefore somewhat fuzzy, in particular on the 10th. But it does not move!

exclaims the reader. Sorry, it does. On the left hand photo the position was and on the 10th. This corresponds to a movement of only to the west and to the south (0.05 mm and 0.07 mm, respectively, on the original plate). You can see it if you measure the distances to the surrounding stars on the figures. From the magnitude it can be estimated that has a diameter of a few hundred kilometers. It is most likely the first known member of a new class of asteroids outside the orbit of Jupiter, and has proposed the name CHIRON (a centaur in Greek mythology). There is, however, still the possibility that it is a comet; at very large distances, it can be very difficult to tell the difference, when no tail shows up and the "head" is perfectly stellar-like.

### Morphological Studies of the Large Magellanic Cloud on ESO Schmidt Plates

This article touches upon a somewhat controversial subject in contemporary astronomy. The structure of the Large Magellanic Cloud is the focus of much research with southern telescopes. Originally classified "irregular", it now appears that it may be possible to break down the LMC into two components, a central ellipsoidal and a somewhat offset spiral structure. The problems of identifying the various stellar components in the LMC, by means of Schmidt plates from La Silla, are discussed. Schmidt telescopes are the most efficient information gathering instruments in optical astronomy. Besides the wide field (up to  $10^\circ$ ) with nearly perfect image definition also at the field edges, the small focal ratio (normally  $f/1$  to  $f/4$ ) permits resolution-limited photographs to be obtained within tolerable exposure times, even on fine-grain emulsions. These advantages are especially useful for the structural study of the Magellanic Clouds. Several ESO Schmidt plates in U-, B-, V-colours of the Large Magellanic Cloud, taken in 1973/74, are used to carry out different studies of the structure of this nearby galaxy and its stellar sub-aggregates. One degree of arc on the sky corresponds to about 1 kpc at the distance of the LMC. Plate-resolution-limited faint stellar images taken with the  $f/3$ ,  $F = 306$  cm ESQ Schmidt telescope have typically diameters of about  $20''$ , which is about 0.3 pc at the LMC's distance. This is the order of magnitude of the geometrical resolution of structural features in the LMC.

### The Structure of the LMC

More than ten years ago, the following picture of the overall structure of the LMC from colour composites of U-, B-, V-, R-photographs with the duplicate of the original Schmidt camera at

the Boyden Observatory was derived: it consists of two components, (a) an extended ellipsoidal galaxy, representing the old stellar population of the bar, and (b) an asymmetric and peculiar Sc-spiral, the centre of which seems to be near the 30 Doradus nebula complex. At least three spiral features can be traced, the most conspicuous one emanates from that centre, crosses the long side of the bar in north-west direction, and splits at its outer part. These spiral features have recently been rediscovered from a study of the distribution of typical spiral tracers like luminous blue stars and HII-regions. A further possibility for a morphological study of the LMC is based on surface photometry, although in principle the interpretation is much more difficult, because integral values along the line of sight are observed. However, photographic isodensity contours from a single Schmidt plate give higher spatial resolution than what is obtained photoelectrically which moreover demands about one hundred times more observing time! Such isodensity contours have been obtained by the Agfa Contourfilm technique. By this simple method, which does not need complex isodensity tracing machines, photographic density differences of about 0.1 or less can easily be separated. Besides the sub-threshold stars, the emission-and reflection-nebulae and the dark cloud areas in the LMC contribute significantly to the isodensity contours. In figure 1 are only shown the less chaotic composites of isodensity contours in the V spectral region, from which figure 2 was obtained by the suppression of smaller details. The outer contour also embraces the OB association of the Shapley constellation. The brightest stars are resolved and do not contribute to the contours. This means that the fainter stars decisively contribute to the brightness distribution within the LMC, although the faint young population I stars no longer can be distinguished from the old population II stars of the elliptical component. The ellipsoidal structure of the bar is clearly recognized from the V-contours. The density levels of the contours were calibrated by star counts in the following way: at positions which appeared undisturbed by interstellar material, the isodensity contour is solely determined by the total number of sub-threshold stars per surface area. They contribute according to the luminosity function. At the relevant positions of the contours, star counts to the limiting magnitudes were made on the two Schmidt plates, reaching absolute

magnitudes of  $M_v \sim -25$  and  $M_v +21$ , respectively. Though the luminosity function is still increasing towards stars of fainter absolute magnitude, those below  $M_v +6$  hardly contribute to the surface brightness. Therefore a correlation should be expected between the average photographic density  $D$  of the corresponding isodensity contour and the counted star number  $N$ . Separation of Populations I and II As mentioned above, the very conspicuous young stellar population I stars and HII-regions, which so clearly outline the spiral features, are no longer distinguishable from the old stellar population II of the LMC below a certain absolute magnitude. How can we then separate the young stars from the old stars in such a faint amorphous substratum? An observational approach for solving this problem is to look at the distribution of those stars, which can easily be recognized, and which exhibit specific features that permit us to classify them as either old or young objects. In the case of population II these are the RR Lyrae variable stars; for population I, we have the A- and F-type Algol eclipsing binaries, which appear to be absent in the population II aggregates of our Galaxy. A search for rapid variables and RR Lyrae stars in the LMC on ESO Schmidt plates is now well under way: I am blinking in a Zeiss comparator a pair of ESO Schmidt B-plates of the LMC, separated in epoch by 1 day. Though the progress is slow because of the enormous surface density of stars, several hundreds of variables have been found on about 25 per cent of the searched plate area. Their amplitudes are between 0.3 and 2 and most of them are apparently fainter than 17. They add to the approximately 2,500 known variables in or in the foreground of the LMC, most of which are brighter than 16. Of course all types of intrinsic and geometric variables with fairly rapid variations contribute to the new sample and no type designations can be given at this moment. However it is known from the recent investigation that the RR Lyrae stars in the LMC have mean apparent B-magnitudes of about 19. A large portion of the detected variables will therefore turn out to be RR Lyrae stars and the rest mainly eclipsing binaries. Another Very Distant Comet Found at ESO 1977 turned out to be a record year for comet discoveries and recoveries. Not less than 20 comets were found and most of the letters of the alphabet had to be used. The present year also got off with a good comet start. Early in January, Dr. P. Wild discovered a 14th magnitude comet with the Schmidt telescope at Zimmerwald and yet another comet was discovered at ESO,

La Silla, on January 12, 1978. Since the ESO comet was reported first, it received the designation 1978a (Comet West) and the Swiss comet is now known as 1978b (periodic comet Wild 2). 1978a was found in the evening of January 12 by Dr. Richard M. West, ESO astronomer, while inspecting plates obtained with the 1 m Schmidt telescope the night before, by night assistant Guido Pizarro. The object was rather faint, magnitude 17 and there was some doubt about the reality. However, another plate the next morning confirmed that it was indeed a comet, slowly moving northwards. It had a rather long tail for a comet of this magnitude, almost ten arcminutes long. Plates were obtained the following nights with the Schmidt telescope and later with the 3.6 m telescope. The orbit has now been computed by Dr. Brian Marsden, who finds that 1978a is very distant; at the time of discovery, it was about 900 million kilometres from the Earth. From eight plates in January it appears that it is moving in a parabolic orbit and passed through perihelion in June 1977 at a distance of approximately 850 million kilometres from the Sun. Thus, 1978a has the third largest known perihelion distance. Two spectra were obtained with the Boller & Chivens spectrograph in the Cassegrain focus of the 3.6 m telescope. To some surprise, it appears that weak emission bands of diatomic carbon (C<sub>2</sub>) may be present, a feature not found in distant comets. Moreover, the tail structure is indicative of the presence of a short tail, in addition to the long dust tail. It is therefore possible that 1978a is "active", even at this large distance from the Sun.

The Helium Variable HD 64740-an X-ray Binary? Professor Kurt Hunger is a frequent user of the coude spectrograph at the 1.5 m telescope on La Silla. His work has mainly concentrated on high-dispersion spectral investigations of stars with the aim of determining their physical parameters and chemical abundances. However, as is sometimes the case in fundamental research, unexpected discoveries may result from other unrelated programmes. The following article is a beautiful example of such an event. Professor Hunger recently left the Technical University in Berlin to succeed Prof. A. Unsold as director of the Kiel Institute. In the course of the spectroscopic investigation of helium-rich stars, carried out at the Institut für Astrophysik of the Technische Universität Berlin and also at the Institut für Theoretische Physik und Sternwarte of the University of Kiel, two of the stars were found to be variable in

the strength of the helium lines: a Ori E as found by K. Hunger, and HD 37776, by S. Clas-Offick. The latter was discovered independently by P.E. Nissen of the University of Aarhus, from narrow-band photometry centered at the line He 1. 4026 Å. This powerful method was later employed by H. Pedersen and B. Thomsen, also from Aarhus, who added a number of new helium variables. Among these, a total of 5 helium-rich are known at present. Despite much effort spent to unravel the nature of the prototype of the helium variables, o Ori E, no satisfactory solution has been found so far. Is it a spotty rotating field star, or is it a close binary with an accretion disc? An argument in favour of the first hypothesis is the recent discovery of a variable magnetic field. The binary hypothesis, on the other hand, is made plausible by the discovery of a variable shell. Whatever final model will emerge, the coming and going of the helium lines must be accompanied by radial velocity shifts that amount to sizeable fractions of the observed rotational velocity,  $v \sin i = 150$  km/s. No Line Shifts However, no shifts are readily detectable. D. Groote, Berlin, and K. Hunger, Kiel, employed a rather sophisticated method to detect radial velocity variations of amplitudes as low as 1 km/s, from 20 Å/mm spectrograms, taken at La Silla. The principle of this method is as follows: two spectrograms taken at different phases are traced on the PDS-Microdensitometer of the Institute in Kiel and the output is stored on magnetic tape. The next step is to bring the two stellar spectrograms to optimum coincidence by shifting in wavelength one spectrogram with respect to the other. This is done in the computer by means of a correlation function that correlates the two spectrograms for the various shifts. The maximum of this function yields the optimum coincidence. The final step then is to find out by how much the laboratory lines are displaced in that given relative position. The result for o Ori E was that no radial velocity variations with amplitudes larger than 2 km/s occur, a fact that poses a serious problem to any model. Spectra of HD 64740 An interesting by-product of the above outlined method was obtained as follows. In order to test the accuracy, the method had to be applied to a star having no radial velocity variations, and resembling as closely as possible the spectrum of o Ori E. These conditions are hardly met by any known stable star. Therefore, the brightest helium star that itself is a helium variable, was chosen, HD 64740, with  $m_V = 4.6$ , and several spectrograms were taken in

rapid succession to ensure that no velocity shifts occurred between the first and the last plate. This test indeed proved the above claimed accuracy. To make further use of these test plates, D. Groote and J.P. Kaufmann, Berlin, started a detailed spectral analysis of HD 64740, based on a computer averaged spectrum that is composed of a total of 8 spectrograms, each belonging to the phase of helium minimum, and each widened to 0.5 mm. The emulsion is Kodak Ila-O, baked in nitrogen. Figure 1 demonstrates how smooth the averaged intensity tracing comes out, the quality almost approaching solar standards! The observed profile of H& and He I 4121 is given by the full Fig. 1. -Computer averaged spectrum of HD 64740 near 4100 Å. Fig. 2. -The spectral region 4650-4700 Å in HD 64740. Note the He II 4686 emission profile with central absorption. Dots represent the theoretical profile obtained from an adapted model atmosphere. The effective temperature turns out to be exceptionally high (27,000 K), for the class of helium variables. At this temperature, the line of He II 4686 Å should appear in absorption with an equivalent width of 50 mÅ. Figure 2 shows a portion of the spectrum from 4650 to 4700 Å. No absorption line is readily detectable at 4686, although the weak lines of O II, N II and C III can be identified down to 20 mÅ. Instead, a broad emission feature is indicated, with a central absorption of the anticipated strength. The emission exceeds the well-defined continuum by 2 percent.

**A New Class of X-ray Sources?** In X-ray binary systems, He II 4686 sometimes appears in emission. HD 64740 indeed is located inside the error box of the weak source 3U 0750-49 as was noted already by Pedersen and Thomsen, whereas the contact binary V Pup, so far suspected to be the candidate, lies 3 arc min outside the error box. Better X-ray positions are needed to confirm the identification. However, if confirmed, it would mean that a new class of X-ray sources has been found. It would also solve the mystery of the helium variables, which would then be binaries containing a compact object, i.e. either white dwarf or neutron star.

**The N119 Complex in the Large Magellanic Cloud** One of the most striking objects seen in blue photographs of the Large Magellanic Cloud (LMC) is a spiral-shaped H II region situated almost at the very center of the so-called "bar" of the LMC. This H II region is generally referred to as N 119, since it is the one hundred and nineteenth entry in a catalogue of emission nebulae in the LMC prepared in 1956 by the American

astronomer Karl Henize. It can also be seen in figure 1 that the area around N 119 appears to be a region of relatively recent and vigorous star formation. Several open clusters may be discerned on the photograph as well as a large number of individual stars which are significantly brighter than the field stars in the LMC bar. In addition, the whole region is covered with faint, diffuse gaseous filaments. What is the nature of this peculiar object? Are the spiral-shaped filaments only the densest parts of a gigantic spherical shell of gas seen projected against the plane of the sky? If so, is this shell expanding? Or are the filaments really thin wisps of gas in the interstellar space? With these questions in mind, and as part of a more general programme investigating the internal kinematics of giant emission nebulae in external galaxies, I have obtained accurate velocity profiles at the positions along the "arms" of N 119.

**The Fabry-Perot Spectrometer**

The instrument used for this work was a photoelectric Pressure-Scanned Fabry-Perot Spectrometer at the 1.5 m telescope of the Cerro-Tololo Interamerican Observatory. The principle of operation of the Fabry-Perot interferometer is illustrated in figure 2. It basically consists of two parallel, semi-transparent mirrors. Parallel light entering the cavity formed by the two mirrors undergoes multiple reflections inside the cavity, producing the interference pattern shown in figure 3. When perfectly monochromatic light is fed into the cavity, it is then concentrated by the interferometer in very narrow rings, each corresponding to the same wavelength, but to a different interference order. Assuming that the mirrors are perfectly parallel, the resolution of the interferometer is given by the width of the rings which in turn depends on the number of reflections inside the mirrors and on their separation. With the advent of low-absorption dielectric multilayer coatings, very narrow rings can be produced. In typical astronomical use the wavelength of the line to be studied is first preselected, usually by means of interference filters. However, the observed light is still not monochromatic and the width of the rings depends also on the intrinsic width of the observed line. Since the "instrumental" width of the rings is very small, very accurate information about the shape of the observed lines can be obtained. Typically, Fabry-Perot interferometers are used in two modes: In the first, more classical mode, a Fabry-Perot is placed in front of a photographic camera, for instance to investigate the kinematics of emission

nebulae. In the second mode, the light from the Fabry-Perot plates is fed into a photomultiplier. The rings are then scanned by changing the length of optical path of the light inside the cavity, either by changing the separation of the plates or by changing the index of refraction of the medium inside the cavity (by increasing the amount of gas between the two mirrors). But what is the advantage of Fabry-Perot interferometers over conventional slit-spectrographs? Well, the resolving power of the Tololo interferometer is about 50,000. To achieve a similar resolving power using conventional coude spectrographs, the entrance slit must be of the order of 0.1 arc-second and with typical seeing conditions of 1 arc-second, only a few percent of the light would actually be used! By contrast, F-P interferometer entrance apertures as big as several minutes of arc can be used without degrading the resolving power. Thus, they are superior for the investigation of the kinematics of extended objects. It should not be forgotten, however, that when using F-P spectrometers only one line can be looked at at the time! The interferometer used in the present investigation works in the pressure mode. The amount of nitrogen gas inside the cavity is continuously increased by a computer-controlled valve while the output of the photomultiplier is read at fixed intervals. The radial velocity of the gas is obtained to an accuracy of about 1 km/sec by comparing the measured nebular profiles with those of a standard hydrogen lamp on the instrument, by using a computer line-fitting programme. Observations of N 119 The results for N 119 are shown in figure 4 where the difference in velocity ( $\Delta V$ ) between the individual positions observed and their mean value (a heliocentric velocity of 276 km/sec) has been plotted as a function of distance to the N119 center projected along the line joining Positions 1 and 9 in figure 1. It is seen that there is a systematic increase in velocity from the southern end of N 119 to the tip of the northern "arm". Is this the consequence of the general rotation of the LMC? The LMC, as a whole, rotates around an axis roughly perpendicular to the (1 to 9) axis of N 119. Therefore, the motions in N 119 ought to reflect those of its parent galaxy. However, in its central regions, the LMC rotates as a solid body with a velocity gradient (along an axis nearly parallel to that of N 119) of about 20 km/sec/deg. Over the observed length of N 119 (7.5 arc-minutes) one expects a velocity difference of only 3 km/sec, i.e. much less than the observed 18

km/sec! The observed velocity field must, therefore, be intrinsic to N 119. A possible explanation for this velocity field is that the arms of N 119 are just the densest parts of an expanding shell. If this were the case, however, one would expect to see a double-peaked profile at the center of N 119 with a separation significantly larger than 18 km/sec, when projection effects are considered. The profiles do not show such a structure, although the resolution of the interferometer is about 9 km/sec. However, the profiles do show a certain asymmetry towards lower velocities. The possibility of expansion cannot, therefore, be entirely discarded.

**The Structure of N 119** We notice in figure 1, that N 119 has a structure somewhat resembling two spherical shells joined at the center of N 119. In fact careful inspection of the photo reveals that N 119 has a "figure 8" shape. But how was this strange structure formed? There are two plausible mechanisms. The first, and perhaps the most classical, is supernova explosions. Here, a star reaches the end of its life and explodes while ejecting large amounts of material at very high velocities. This material then sweeps out the surrounding interstellar gas and is decelerated by what could be called interstellar "friction", forming gigantic loops. An alternative and very attractive mechanism has often been invoked in recent years. Bright supergiant stars (such as Wolf-Rayet stars) are known to lose large amounts of mass from their atmospheres at velocities reaching thousands of kilometers per second. These so-called "stellar winds" act upon the interstellar medium more or less like a supernova blast, producing what has been called "an interstellar bubble". Since a stellar wind continuously drives the bubble outwards, while a supernova blast gives it only one huge energetic push, there are certain physical differences between the two mechanisms which in principle might allow us to distinguish between the two possible origins for the observed bubbles. This, however, is not a simple problem and it has been the subject of much research during the past few years, especially in connection with the LMC. In the case of N119 it is known that it contains at least one very bright supergiant star, located right at its center. This star, called S Doradus, has been intensively studied. S Doradus could be driving a massive wind, but it is not easy to explain how it could produce a structure like that of N 119. On the other hand, radio observations of N 119 do not show that

a supernova explosion has recently taken place near the nebula. Clearly, a detailed study of the velocity field of N 119 would be of much help to understand the nature of this interesting nebula. The photographic Fabry-Perot interferometer used at La Silla would be an ideal instrument for this investigation. Together with accurate radial velocity information, this instrument provides the necessary spatial resolution required to properly map the velocity field around N 119.

How Stars are Born There is a vivid interest among astronomers in the early phases of star formation. In the last issue of the Messenger a catalogue of stellar birth places was introduced. The present article discusses radio, infrared and optical observations of a particularly interesting object. In recent years both radio and infrared astronomy have revealed details of the dusty environment of star formation. These regions are characterized by the presence of "Compact H II regions", which are often associated with HO and type I OH masers. Infrared sources (1 to 30mim) are often seen in or nearby the compact H II regions and sometimes combinations of these sources can be found close to visible H II regions.

The "Cocoon" Model These regions can best be explained quantitatively by the recent models of Kahn and Cochran and Ostriker, who propose the following scenario: A protostar forms by accretion in a dusty interstellar cloud. As the star's luminosity increases with time, the accretion is halted by radiation pressure. A dusty "cocoon" remains, within which is a smaller ionized zone surrounding the star. In this phase the dust and gas are competing for stellar photons and a situation can arise where the dust is heated to a higher temperature than the surrounding gas and can give rise to the necessary infrared radiation capable of pumping the HO and OH masers. As the star evolves the cocoon fragments and the compact H II region become visible. At a later stage, as the star settles into the Main Sequence and the dust shell dissipates further, a conventional Stromgren sphere (H II region) may become visible. This later stage may be exemplified by regions such as S888 or Sharpless 2-106 where visible Ha emission is seen. Here the infrared source may be interpreted as an O star with high visual extinction due to dust.

Observations at Westerbork and La Silla To investigate the various phases and accompanying physics of these regions, a programme to study regions of star formation using the ESO infrared equipment at the 1 m ESO telescope at La Silla

has been instigated. First results of this programme are shown in Figures 1 and 2. In Figure 1 the radio brightness contours are shown for the source G12.2-0.1. These observations were made with the Westerbork Synthesis Radio Telescope at a wavelength of 6 cm; the beam size was 6x31 arcseconds. The compact H II region is indicated by A and two other radio bright regions are indicated by B and C. Although the Westerbork beam is elongated at this low declination, the radio contours can still be seen to trace out a shell-like structure. We have marked also the positions of the OH source and HO sources. Subsequent mapping of the region at 2.2mi at La Silla using a 10 arcsec diaphragm and 37 arcsec chop on the sky revealed 3 infrared sources. Two are shown in Figure 1, indicated as IRS 1 and 2; the third was detected in the reference beam and is just outside Figure 1. Of these sources, IRS 1 is the most interesting, coinciding with the compact H II region. Recent position measurements of the HO source put component A, IRS 1, and the HO source within 2 arcsec or 0.03 pc-an unusually close association. We have measured the spectrum of IRS 1 from 1 to 5mi. A Magic Eye for Astronomical Spectrophotometry

The ability of a telescope to detect faint celestial objects not only depends on the linear size of the telescope, but also upon the efficiency of the light detectors that are used to register the light. For many years, most astronomical spectra were obtained on photographic plates. However, even the best of these rarely achieve detective quantum efficiencies above a few per cent, i.e. they only "catch" two or three out of every one hundred photons hitting the emulsion. During the past decade much effort has therefore been concentrated in astronomy on how to improve the detector efficiency in order to make small telescopes "larger" and large telescopes "very large". For instance, a telescope with a mirror diameter of one metre and a detector efficiency of 50 per cent is (for many astronomical applications) equivalent to a 5 metre telescope with a 2 per cent detector. In this article, discusses one of the new detectors, the RETICON array. The use of large enough beam apertures, sometimes larger than the diffraction limit, with current mm telescopes is discouraged by the dramatic increase of the sky noise with respect to diffraction-limited observations. The alternative is to obtain information on the spatial distribution of light emission at submm-mm wavelengths by suitably mapping some bright nearby sources. This will be the goal of our next observing

runs at La Silla. Let us briefly discuss currently available information on the subject. A direct comparison of optical and far-IR profiles has been done on three nearby edge-on spirals using the IRAS CPC instrument. Unfortunately, edge-on galaxies do not allow a detailed study of the radial disk structure. However, a comparison between the far-IR emission along the major axis can be performed with the optical old-disk light. From their study of NGC 891 it seems that the 100- $\mu\text{m}$  emission is more extended than the 50- $\mu\text{m}$  one. They suggest that the cold diffuse interstellar component dominates with respect to the optical emission at distances beyond 9 kpc from the centre. For the other two objects similar far-IR and optical light profiles can be inferred from these observations. This seems to indicate that the cold dust emission at mm wavelengths might be quite extended with respect to the warm component and the optical emission, although IRAS maps at large radii are too noisy for any definitive conclusion to be drawn. For a sample of large galaxies partially resolved by IRAS the mean ratio of the far-IR (60  $\mu\text{m}$ ) to blue-light isophotal diameters turns out to be  $0.98 \pm 0.25$ , which means that on average galaxies have far-IR extensions comparable to their optical sizes, quite in agreement with previously mentioned results. In this case, however, the observed mean of the ratio of the effective far-IR aperture diameter (which includes half of the galaxy's light) to the isophotal radius for 11 objects of this sample turns out to be almost half of that of the blue light. This difference indicates that the IR emission could be more centrally concentrated than that of the blue light. A more centrally concentrated mm emission with respect to the optical may be due to the effect of extinction on the blue radiation toward the central regions of the galaxies. This indication agrees with recent reinterpretations of the optical galaxy profiles which seem to show non-negligible light absorptions in the galaxy cores.

**Conclusions** Our knowledge of galaxy spectra in the submm band is still subject to relevant uncertainties. Should galaxy sizes at such wavelengths be comparable, or even larger, than those in the optical light, then mm emission and the corresponding amount of cold dust in the interstellar material would be significantly larger than expected. Detailed observations are planned to clarify this issue. Several important consequences can be envisaged. Since the millimetric

flux is proportional to the dust mass emitting at these energies, the amount of cold material in galaxies could have been underestimated. This fact could lower the gas-to-dust ratio claimed for spirals from CO and far-IR measurements, to values comparable to that observed in the ISM of the Galaxy. The contribution of discrete sources to the fluctuations of the Cosmic Microwave Background at small and intermediate scales is strongly sensitive to the galaxy spectra in the long wavelength spectral domain. An enhanced thermal dust emission from normal galaxies with respect to current estimates would eventually prevent detections of any intrinsic anisotropies of the CMB. More precise definitions of galaxy spectral energy distributions and local luminosity function would allow to improve the estimates of number counts and contributions to the diffuse background. Observations by FIRAS on COBE would eventually detect such a background.

THE ROSETTA NEBULA The "Rosetta Nebula" is situated just north of the celestial equator, in the constellation of Monoceros (the Unicorn). In its middle is the stellar cluster NGC 2244, one of the youngest of its kind known. The distance to the nebula and the cluster is about 4000-5000 light-years. There is little doubt that the young stars were born in the Rosetta Nebula and have only recently become visible. This is because they have blown away the gas and dust from their immediate surroundings. The Rosetta Nebula displays a number of dark lanes which are caused by the shadowing effect of dust clouds. Its red color is caused by the light emission of hydrogen atoms and the different color hues reflect local variations in the temperature and composition of the nebula. This photo is a composite from three black-and-white photos obtained with the ESO 1-m Schmidt telescope at La Silla.

New ESQ Preprints Scientific Preprints Deep Luminosity Functions of Globular Clusters. IV. NGC 6171. Monthly Notices of the Royal Astronomical Society. Reduced Upper Limits on the Equivalent Width of Interstellar Li 1 670.8 Towards SN 1987A. Astronomy and Astrophysics. Post-Perihelion Observations of P/ Halley. III. An Outburst at  $R = 14.3$  AU. Astronomy and Astrophysics. Core Overshooting and Stellar Evolution. Astronomy and Astrophysics Letters. Distortions of the CMB Spectrum by Distant Clusters of Galaxies. Astronomy and Astrophysics Letters. Dissipationless Collapse and the Shape of Isophotes. Monthly Notices of the Royal Astronomical Society. Abundance

Gradients and Galaxy Formation. Astrophysical Journal Letters. NTT Interstellar NaI Observations of the Two Faint Optical Companions of SN 1987A. Astronomy and Astrophysics. The Origin of the X-Ray Background. To be published in the proceedings of the 28th Yamada Conference on "Frontiers of X-Ray Astronomy". Dust and Line Luminosities in SN 1987A. A Theoretical Investigation of Population II Red Giant Clumps. Astronomy and Astrophysics. The Clustering of Quasars and its Evolution. To appear in PASP Conference Series. The Evolution of Carbon Stars in the Magellanic Clouds. Astronomy and Astrophysics Supplement Series. The Ultraviolet Spectrum of NGC 4151 from 1978 to 1990. Astrophysical Journal. On the Mass of the Compact Object in SS 433. Nature. Luminosity Functions, Relativistic Beaming, and Unified Theories of AGN. To appear in the proceedings of "Physics of Active Galactic Nuclei". Fanaroff-Riley I Galaxies as the Parent Population of BL Lacertae Objects. Radio Constraints. Astrophysical Journal. Observational Aspects of Stellar Seismology. Technical Preprints The Temperature Structure Function for Complex Terrain. Paper presented at the Eighth Symposium on Turbulent Shear Flows, Technical University of Munich, September 9-11, 1991. The Use of Differential Adaptive Optics for Astronomical Interferometry. On the Optimization of Partial Adaptive Optics. Submitted as Technical Notes to Applied Optics. Solar Feature Correlation Tracker. To appear in SPIE Proceedings: Technical Conference 1542 "Active and Adaptive Optical Systems", San Diego, July 22-24, 1991. Adaptive Optics System Tests at the ESO 3.6-m Telescope. To be published in Proceedings of SPIE, Vol. 1542. Adaptive Optics for the European Very Large Telescope. To be published in Proceedings of SPIE, Vol. 1542. Latest Developments of Active Optics of the ESO NTT and the Implications for the ESO VLT. To be published in Proceedings of SPIE, Vol. 1542. Giotto to Visit Comet P/Grigg Skjellerup in 1992 The Mission Background Shortly after its successful encounter with comet P/Halley on March 14, 1986, which provided a wealth of exciting new data, the GIOTTO spacecraft was put into hibernation with the prospect to reactivate it later on and to retarget it for an encounter with comet P/Grigg Skjellerup in 1992. After nearly four years in hibernation the spacecraft was reactivated on February 19, 1990. Subsequently, the check-out of the spacecraft and the scientific payload demonstrated that GIOTTO is in a state to

support such an extended mission and that a complement of at least six of the original eleven science instruments was functional and able to provide data of high quality during a second cometary encounter. Science investigation, which can be made with this payload, include: characterization of the changing features of the solar wind flow and observation of cometary pick-up ions and anomalous acceleration -determination of electron densities -observation of upstream waves, determination of the location of the various boundaries (bow shock, ionopause, cometopause, etc.) -observation of the magnetic pile-up region and cavity -determination of the dust spatial density and size distribution and of the optical properties of the dust grains -discrete gaseous emissions -combined dust and gas densities Despite the fact that the Halley Multicolour Camera (HMC) will not provide any images of the nucleus -the aperture of the HMC is blocked most probably by a piece of the outer straylight baffle that has been severely damaged during the close encounter with P/Halley -and that the Neutral Mass Spectrometer had all detectors damaged, ESA's Science Programme Committee confirmed the high scientific value of a GIOTTO Extended Mission at its meeting on June 12 and 13, 1991. The Orbit With an aphelion distance of 4.94 AU comet P/Grigg Skjellerup belongs to the Jupiter family of comets. The perihelion distance of 0.99 AU -together with other characteristics -makes this comet very attractive for an encounter with GIOTTO on July 10, 1992, just 12 days before perihelion passage of the comet. Table 1 contains the most recent orbital parameters of P/Grigg Skjellerup determined at the European Space Operations Centre ESOC in Darmstadt. The times are ephemeris times ET, the angles are referred to the ecliptic of mean equinox 1950.0 and the definition of the non-gravitational parameters A1 and A2 is that of Marsden et al. As can be seen from Table 1, the non-gravitational forces are found to be small and perturb the cometary motion only slightly. The Visibility Before the Encounter For the forthcoming perihelion passage the visibility of P/Grigg Skjellerup from Earth starts in July/August 1991 (morning sky). The comet can be observed from both hemispheres of the Earth up to about May 1992. It will stay relatively close to the celestial equator (between +13 and -6 deg declination). The number of dark hours per night is higher for observers located in the northern hemisphere until April 1992. Though the visual brightness of

the comet may be rather faint, measurements of astrometric positions and physical properties of the comet can certainly be obtained by large telescopes during this time interval. Within two months before the Giotto encounter with P/Grigg Skjellerup the comet will be observable from the southern hemisphere only. The brightness and activity of the comet will increase towards perihelion passage. However, intensive observations may suffer from both the rather moderate total coma brightness (about 13 mag in the maximum) and the small elongation of the comet from the Sun which will result in just one to two hours observing time at low elevation in the evening sky. However, scientific observations, both astrometric and astrophysical ones, collected during this time interval, will be of high importance for the success of the fly-by and the interpretation of the GEM data. The Light Curve For most apparitions the visual brightness of P/Grigg Skjellerup was estimated during about a two-month time interval before and after perihelion passage. Only very little is known about the light curve outside this arc of the orbit. Some puzzling measurements of the nucleus' brightness are available from the 1986/87 apparition of the comet. The total and the nucleus light curve of comet P/Grigg Skjellerup is plotted in Figure 2. The data were calculated from the light curve coefficients published by Nakano and Green. From the beginning of 1992 until the Giotto encounter the nucleus brightness should increase only slightly from about 20.5 to 19 mag. For the total coma brightness a maximum value of about 13 mag can be expected around perihelion passage which is close to the Giotto fly-by. The light curve of Nakano and Green indicates that the comet may show a rather late onset of significant coma development and a steep brightness increase pre-perihelion. However, this prediction of the early coma activity may be prone to errors because of the fragmented observational coverage of this phase during previous apparitions. This trajectory change Giotto was again put into hibernation. It will be reactivated in May 1992 for the preparation of the comet fly-by. Based on the present knowledge of the Giotto and P/Grigg Skjellerup orbits the Giotto fly-by at the comet will take place on July 10, 1992, 15:25  $\pm$  00:10 UT at 1.01 AU distance from the Sun and 1.43 AU distance from the Earth. The relative fly-by velocity will be 13.99 km/s. Figure 3 sketches the pre-encounter approach of both objects together with the respective

position of The Fly-by of GIOTTO After the encounter with comet P/Halley on March 14, 1986, the GIOTTO spacecraft was put into hibernation, an operations state of minimal on-board activities without control from Earth. In February 1990 GIOTTO was reactivated by ESOC for the first ever swing-by of an interplanetary space probe at Earth. The swing-by was successfully performed on July 2, 1990. GIOTTO was redirected to an encounter with comet P/Grigg Skjellerup in July 1992. The final targeting of GIOTTO for the fly-by will be performed by ESOC about 2 days before encounter at the latest. However, in any case GIOTTO will pass the orbital plane of the comet from North to South during the encounter. A Request for Astrometry Comet P/Grigg Skjellerup was observed during all 13 apparitions since 1922. Because the 1987 apparition was the last one before the GIOTTO swing-by at the Earth on July 2, 1990, which redirected the satellite towards P/Grigg Skjellerup, ESOC has instigated an appeal for astrometric measurements of the comet to the observers. Altogether 60 accurate position measurements were obtained -more than double the number collected in 1982 and several times as many as at most other apparitions. In 1992, high-quality astrometry of P/Grigg Skjellerup is needed at ESOC for the final targeting of GIOTTO towards an optimum fly-by at the comet on July 10, 1992. The importance of ground-based astrometry for the orbit determination of P/Grigg Skjellerup is even higher than during the Halley campaign in 1985/86 since for GIOTTO there will be no Pathfinder Project using observations from other cometary missions. Therefore, ESOC is seeking the direct collaboration with professional observers who are able and willing to obtain astrometric positions of P/Grigg Skjellerup in 1991/92 before the encounter. The observers should be able to communicate the astrometric positions of the comet to ESOC within a few hours to two days after observations in order to allow an immediate update of the cometary orbit for the fly-by planning of GEM. All positions of the comet reaching ESOC before July 8, 1992, will be considered in the planning for the GIOTTO encounter. However, of highest priority for the fly-by targeting are position measurements of P/Grigg Skjellerup obtained within two months before the encounter. During this period the comet will be best observable from the southern hemisphere though it might be a difficult task since the comet will be faint and close to the horizon. Observations are being planned at ESQ La Silla, in collaboration with R. West,

who also observed Halley in 1986. For the improvement of the orbit determination accuracy so far unpublished position measurements of the comet obtained during previous apparitions are very welcome at ESOC. Astronomers who are interested to contribute to the P/Grigg Skjellerup astrometry campaign for the GIOTTO fly-by may contact for further information: ESOC Flight Dynamics. Data Analysis Workshop This volume contains 236 pages and is offered at a price of DM 30.- (packing and surface mail included). Payments have to be made to the ESO bank account 2102002 with Commerzbank Munchen or by cheque, addressed to the attention of Please do not forget to indicate your full address and the title(s) of the Proceedings. Other ESO publications recently published are: VLT Report No. 63: "Field and Pupil Rotations for the VLT 8-m Unit Telescopes". VLT Report No. 64: "VLT Combined Focus Efficiency with Optical Fibres". ESO Technical Report No. 15: "A Study of the Potential of Heterodyned Holographic Spectrometry for Application in Astronomy". This particular kind of stony-iron meteorite is known from about 30 other locations only. The amount recovered at Vaca Muerta has tripled the material available to laboratory study. When fully analysed, the seventy-seven "dead cows" from Atacama will undoubtedly provide us with much new insight into the enigmatic history of the early solar system. GPO Observations of a Geostationary TV-Satellite Quartet Introduction During an astrometry campaign on asteroids a test was performed to observe geostationary satellites with the GPO 40-cm astrograph at ESO La Silla. In several nights from April 10 to 22, 1991, spacecraft located at geostationary longitude 19 deg West about 35800 km above the equator were photographed with the GPO. At this longitude in total four telecommunication satellites are operated by three different control centres, i.e. TDF-1 and TDF-2 by CNES Toulouse, OLYMPUS by Telespazio Fucino under contract of ESA and TV-SAT-2 by GSOC/DLR Oberpfaffenhofen. In order to guarantee the contact with the fixed mounted user antennas on ground, the individual spacecraft have to be kept in control boxes of certain latitude and longitude intervals. The respective control boxes are in inclination below 0.1 deg for all four satellites, in longitude within 18.7 to 18.9 deg West for both TDF satellites, within 18.9 to 19.1 deg West for OLYMPUS and within 19.1 to 19.3 deg West for TV-SAT-2. A violation of these so-called deadbands by one of the satellites

may not only cause problems for the users on Earth because of fading signal strength, but may also be a risk for the spacecraft themselves in particular if the longitude window of a neighbouring satellite is entered and a collision of both satellites may occur. Since perturbations from the Earth, the Moon and the Sun cause a geostationary satellite to drift away from its nominal position, the control centres have to correct the orbit regularly both in inclination and in longitude by so-called "station-keeping manoeuvres".

The GPO Observations The geostationary longitude 19 deg West over the Earth equator compares to the telescope coordinates of +4.6 deg in declination and hour angle 3 h 52.8 min East for the La Silla observatory. During the about 30-minute time interval of dark-room work per night (in order to change plates for the asteroid programme) old ORWO ZU21 plates found in the refrigerator of the telescope building were exposed with the tele-stationary satellites like the 19 deg West quartet are usually operated with perigee pointing approximately towards the Sun. Therefore, the daily longitudinal and radial libration is synchronized, which can clearly be seen in Figure 1. All four spacecraft moved from East towards West during the observing interval. This picture also illustrates the daily brightness variations of the three-axis stabilized satellites because of the temporal changes in the phase angle Sun-spacecraft-observer. All four satellites are brightest around 1 UT (second exposure), i.e. when the Sun is in "opposition" to the satellites. Furthermore, the scope pointing to the geostationary satellites. The telescope tracking was switched off during these satellite exposures. Therefore, on the plates the stars appear as dark lines across the field of view, while the geostationary objects are black dots or short trails. Since the satellite orbits have an inclination usually different from zero and the orbital eccentricity is not exactly zero, the geostationary spacecraft perform a daily libration which is a superposition of a latitudinal, a longitudinal and a radial motion similar to 3-dimensional Lissajous figures. A part of the daily libration of the four satellites is visualized in Figure 1 in projection onto the celestial sphere. This image obtained on April 22, 1991, contains in total 11 five-minute exposures with starting time each full hour. However, only the first nine exposures show the geostationary quartet on the plate. Unfortunately, the telescope moved slightly in declination for a yet unknown reason between exposure 4 and 5 which has introduced an artificial shift of the

satellites' motion towards south. For a minimization of fuel consumption geo-exposures 10 and 11 obtained at 9 and 9.30 UT, respectively, did not depict the spacecraft at all, probably because most of the satellite surfaces seen from La Silla were in shadow. The Deadband Violation of OLYMPUS In the sequence of exposures obtained between April 17 and 20, 1991, a so-called longitude deadband violation of OLYMPUS was documented by chance. It occurred because of altitude control problems due to onboard sensor errors. For comparison on April 10 and 11, 1991, all four spacecraft were positioned within their control boxes. On April 17, 1991, OLYMPUS was already about to pass its eastern longitude limit. It entered the neighbouring TDF control box and can be seen between TDF-1 and TDF-2 on April 18, 1991, and even east of both TDFs on April 19, 1991. The following day it drifted back to its nominal control box where it was found inside again on April 22, 1991. Fortunately, this "excursion" of OLYMPUS out of its control box has caused no hazard to the neighbouring TDFs. However, the control centres certainly try to avoid such contingencies during normal operations, in particular for colocated spacecraft like the quartet at 19 deg West. The astrometric plates obtained on the geostationary satellites can be used to measure the angular distances between the spacecraft. These angles can be transformed into projected inter-satellite distances in kilometers with an accuracy of less than 1 km in geostationary orbit. This is at least of the order or even better than the accuracy of the proximity checks using radio tracking data. Therefore, the optical observations may be useful as an additional check for the orbital proximity calculations of colocated geostationary satellites.

Mini-Workshop on Large-Size CCDs at ESQ

Which is the maximum size of a high-quality scientific CCD detector that industry can now deliver? How can the UV-blue quantum efficiency of the devices be enhanced? What are the best design approaches and the limiting performance of CCD controllers? These questions are puzzling engineers and scientists who have to do with the definition, design and procurement of the detectors and their control systems for astronomical applications. Projects like the instruments for the ESO Very Large Telescope stress the need of devices of large size and state-of-the-art performance in order to take full advantage of the larger collecting area of the telescopes. To focus on these open questions and to obtain a snapshot of this fast developing field, ESO organized on June 18th and 19th

in Garching a mini-workshop on "Large Size CCD". Invited were representatives from European groups with a proven experience in this field, a few experts from overseas and speakers of CCD manufacturers with an interest in the astronomy market. The workshop was organized in three sections dedicated to CCD Controllers, CCD Operation/Testing/Design and finally to Presentations from industry. The workshop programme gives titles and authors of the talks whereas the paragraphs below summarize status and highlights of the various topics, as seen through the biased eyes of the authors. A variety of systems have been developed at different observatories with the aim of optimizing those operating parameters which are of relevance for the astronomical applications. The analog section of the controller is beside the intrinsic quality of the CCD the dominating part as for what concerns the final quality of the signal processing and the CCD scientific data. The intrinsic CCD read-out noise has been significantly improved in the last years due to the progress in the semiconductor technology and this development has forced a requirement on the analog chain to be based on circuitry design with a 1 electron equivalent noise. Overall system noise levels of 3-4 electrons have now been reached in optimal operating conditions at various observatories, approaching with CCDs the performance of photon-counting detectors. In spite of various developments concerning analog drivers and video circuitry, there has been a relative standardization of the principles of the video signal processing. Improvements are still to be expected by the use of improved components and carefully tuning of the analog modules. Concerning the digital controller section the current systems are routinely operating at a good level of reliability. Main disadvantages are their hardware complexity and a high power consumption, besides a great variety of sometimes not standardized interfaces to the host computer. Facing the challenge of telescope arrays like ESO's VLT, the reliability of each CCD system has to be improved drastically in order to ensure the operation of multiple systems at the various foci and telescopes. A way to reach this aim and to improve at the same time many other features of the CCD control system is the use of Digital Signal Processors (DSPs) and Transputers (TPs) which have become commercially available during the last few years. Basically being a special species of microprocessors, TPs offer an increased capability of standard links to the

outside world, e.g. the host computers, a low power consumption and less complexity of the surrounding digital circuitry. Moreover, they can also be used to collect and optionally preprocess large amounts of digital video data. The higher flexibility simplifies the implementation of multiple windows/binning, CCD mosaic operation, and total software control of all static and dynamic CCD voltages. These integrated devices are now established from different companies with standardized interfaces and seem to be well suited in leading to very small, lightweight and reliable systems possibly attached to the CCD cryostats. Several observatories have now controllers based on Transputers at various stages of development. Unlike the other groups which are using TPs combined with conventional digital circuitry, ESO currently investigates the use of a combined system of TPs and DSPs. This system aims at combining the interface advantages of TPs and the quick, extremely accurate timing pattern generation of DSPs needed for CCDs. In the section dedicated to CCD OPERATION/TESTING/DESIGN, the presentations provided updated information on two areas of crucial importance. Speaking about "Custom Design of CCDs for Astronomy", it was pointed out that in the past the design of astronomical instruments had always to adapt to the light-sensitive area format and pixel size of commercially available CCD detectors. The development of custom-designed CCDs with guaranteed performance by the industry results in relatively high costs. Lack of the sophisticated design equipment needed for the work made it impossible for the customers to contribute in an active way to the CCD layout work. Through a collaboration with a manufacturer a new approach for CCD custom design has now been successfully made, where customer and industry share part of the design and test work as well as the fabrication risk. Due to the repetitive structure of CCD light sensitive areas, the design work of the actual CCD semiconductor layout can be done for a single fraction of the CCD structure by the potential user himself on a commonly used IBM compatible PC with Autocad Software. During this design work it is possible to create or to borrow all the unique structures of the CCD (output amps, bus inputs, pixel cell). Useful is here the cooperation among potential users with their own design libraries. The resulting file is then transferred to the CCD manufacturer now acting more like a simple CCD foundry, who can edit the file with a standard text editor and

introduces some replication factors for the repetitive parts (pixels, phase structures). The file (now containing the whole CCD structure) is finally converted into a CAD File for lithography with reticle masks. After the CCD production on the wafers the testing effort is again shared between the customer and the manufacturer, with the latter doing the basic on-wafer electrical testing while the first carries out the time-consuming image and full performance tests. When the wafer quality is high and the manufacturing process clean and well tested, one wafer run can produce a relatively large amount of useful devices at a low cost and with a reasonable manpower investment. A few runs of custom-designed wafers have been successfully completed at a manufacturer. The CCD group coordinated by Johannes Andersen has shared the costs of one of these runs and is expected to receive the first CCDs by the end of the year. An example of a custom designed CCD wafer is shown in Figure 1. What is the major drawback of custom designed CCDs? With all their interesting advantages, they remain front illuminated devices with no sensitivity in the astronomically very important UV-blue spectral region. In his second talk at the workshop, the problem of CCD thinning and the companies/laboratories who have acquired experience in this field was discussed. Various mechanical/chemical processes of whole wafer or single cut CCD thinning have been used with different degrees of success. A manufacturer seems at present to be the only one who regularly masters the process on devices as large as 50 x 50 mm. Thinning experiments are also carried out at the CCD Lab of a research center. These are potential addresses for CCD customers with a batch of printed wafers to thin, but the rate of success and the cost of the process are still not well defined. The final session was dedicated to the PRESENTATIONS FROM INDUSTRY. The participants had the opportunity to hear reports on the latest products from most of the major manufacturers of scientific CCDs. On the size of CCDs, the trend is clearly dominated by the need to maximize the number of devices which can be fitted on 10 cm wafers to reduce the costs. This explains the success of the 2048, 15-mim-pixel format which allows to squeeze 4 devices on one wafer. Improved performance was also reported due to the progress in microchannel technology for a high CTE at very low light levels and the Lightly Doped Drain (LDD) on-chip preamplifier for lower noise and higher

gain. On the performance of CCDs that a potential customer can get now off-the-shelves of industry, we suggest to contact the commercial agents of the companies. As a provisional guideline, we give below a summary of the situation as obtained from our notes on the presentations and the data sheets distributed at the workshop. A manufacturer has obtained good results with 1,200 x 400, 27-mim-pixel thinned devices and is currently redesigning the chip support for improved flatness. A manufacturer has produced thinned devices in the 512 format and has made plans to develop a 2048, 17-mim-pixel, 3-edge-butable, thinned device. A manufacturer announced the availability of their 2048, 24-mim-pixel, thinned, CCD for July 1991. The device is offered with a guaranteed performance and it appears the only one of this size available on short term. The price was not named at the workshop, but the manufacturer is now willing to quote it. A manufacturer presented their results on CCDs of different formats, like the 2048, 15-mim-pixel and the 4096, 7.5-mim devices. A manufacturer now delivers front-illuminated devices with up to 1242 x 1152 pixels (22-mim size) and has announced a 2186 x 1152 pixel CCD, butable on two sides. They also obtained good results in thinning smaller devices. Beside the presentations, the workshop offered many opportunities to discuss the various topics, and to share know-how and expertise. We have the feeling that the meeting fully met its main goals, providing an updating view of the subject and favouring the collaboration between the different groups and with the industrial suppliers.

### The Use of Photography in Astronomy: Some Thoughts about Schmidt Telescope Wide-Field Work

Introduction During the past decade, a revolution has taken place in the field of astronomical detector techniques, and astronomers all over the world are profiting from new digital devices like CCDs and photon-counting detectors. Indeed, it might appear that photography, previously so widely used in astronomy, will soon be a thing of the past in professional work, no longer of any value in front-line investigations. Among the various types of ground-based astronomical instruments now in use, none has been more intimately connected to the development of improved photographic methods over the past decades than the large Schmidt telescopes. The future of the photographic detector in astronomy is therefore largely dependent on its useful application at Schmidt telescopes. In this article I shall explore a

question that is now being posed at many observatories: is it reasonable to continue to use photographic plates in Schmidt telescopes or has the time come to aim at a rapid implementation of CCD techniques, also here? In this connection, it should be remembered that "an astronomical instrument is only as good as the scientific results it produces". This maxim is not always kept in mind and around the world there are quite a few examples of "senior" telescopes with associated methodology which for sentimental or traditional reasons continue to be in use, long after it has been shown that their ability to produce good science of current relevance has begun to decline. Moreover, it is not always fully appreciated that it is the science resulting from an instrument that defines its current value in a global context, not the technology it employs. In addition to the fundamental limitations imposed by the funds available in any one place, there may of course be other, perfectly valid reasons, for instance long-term continuity of observational programmes and the availability of well-established calibration systems, which makes it advantageous to keep using certain telescopes, auxiliary instruments and associated methods, also well beyond their prime years. However, in the ideal case, and certainly for most front-line research programmes, it is clear that failure to adopt new and improved technology and observational methods will eventually result in slower scientific progress and ultimately, in falling behind the rest of the world. It is also a question of the real cost of the observational data -if from a certain moment a new method allows the same astronomical information to be gathered at significantly lower expense in time and manpower, then a change-over ought to be seriously considered. The question above can then be reformulated: would the introduction of CCD techniques in Schmidt telescopes enhance their ability to produce good science? The most important intrinsic properties of the photographic and CCD techniques are compared in Table 1. Both types of detector demand very careful handling. To ensure the best use of CCDs, it is necessary to obtain bias, flat-field and dark frames, and the elimination of artificial and natural blemishes, e.g. cosmic-ray events, implies time-consuming work which is not necessarily "easier", but perhaps better controllable than the delicate hypersensitization and processing of photographic plates. In this connection, it is an interesting fact that only the photographic emulsion is at

the same time detector and storage medium and that therefore the important question of complete archiving of the raw observational data for the benefit of future users is a priori taken care of in photography. On the other hand, the digital output from CCDs is perfectly suited for direct input to computers, thereby streamlining the reduction procedures and, above all, eliminating the need for a preceding transfer from analogue density to digital numbers via more or less noisy registration devices. It is therefore clear that the main advantages of the photographic plate over the CCD are its larger size and the ease with which the observed data can be archived. Within the foreseeable future, no CCDs, not even mosaics, can be expected to approach the field size and thereby also the total information storage capacity of large photographic plates. But are these advantages important enough to warrant further use of the photographic method in existing large Schmidt telescopes? Would it not, with the shorter exposure times needed, be possible to cover the same total sky field in the same time with multiple CCD exposures? And why should we continue to use photographic methods at large Schmidt telescopes, when the digital detectors are so much more accurate and "clean"? As we shall see below, a more detailed analysis indicates that the photographic technique is still superior for wide-field, high angular resolution Schmidt work.

### Current Use of Schmidt Telescopes

All of the world's large Schmidt telescopes (e.g. with aperture above 50 cm) currently rely on the use of large photographic plates, which permit the simultaneous registration of direct images or slitless (objective prism) spectra of objects in sky fields of the order of  $5^\circ \times 5^\circ$  or larger, with angular resolutions that in most cases are equal to the instantaneous local seeing. Schmidt plates are used for a great variety of astronomical programmes. Some of these do not fully take advantage of the large field which is the main virtue of Schmidt telescopes, and could well be carried out with other instruments. There are in the literature many examples of research programmes with photographic Schmidt telescopes which are concerned with individual objects whose small angular dimensions would make them fit into a normal CCD frame. In these cases, it would clearly have been more efficient to use the CCD technique at another telescope, although such an instrument may not have been accessible to the astronomers concerned. "Narrow"-field Schmidt work will not be further

considered here and in what follows, I shall restrict myself to sky surveys (coverage of a major part of the sky, often in several colours) and sky patrols (repeated coverage of the same part of the sky). Schmidt plates, whether obtained for large sky surveys or for less comprehensive programmes, and whether originals or high-fidelity copies, serve several scientific purposes, of which the following are particularly important: Record for later research (e.g. to check the previous behaviour of an object of current interest) Inventory of objects in a particular sky area (e.g. to produce catalogues of all objects with particular characteristics) (Optical) identification of objects (radio, X-ray, gamma-ray) Serendipitous discoveries (e.g. of minor planets, comets and variable objects) Astronomical observations, including those made with Schmidt telescopes, provide different types of information (data) which may be classified into a number of natural areas (not necessarily in order of perceived significance):

1. Discovery of new objects
2. Astrometric positions of individual objects
3. Photometric intensities of the radiation received from individual objects
4. Detection and characterization of temporal variations in position and intensity of individual objects
5. Identification of larger structures and their surface characteristics, including higher-order clustering of individual objects.

How well is the wide-field Schmidt photographic technique doing in these areas? Discoveries. There is little doubt that Schmidt surveys in different colours, direct or through an objective prism, as well as Schmidt patrols, constitute one of the most successful techniques of identifying (i.e. discovering) particularly "interesting" new objects, due to the ability to cover large sky fields at high angular resolution and at regular time intervals. A well-known example is the Palomar 48-in Schmidt (Oschin) telescope with which large quantities of peculiar objects have been identified for subsequent, detailed studies with the 5-m Hale telescope. Most new minor planets and a large fraction of the new comets are still found on Schmidt plates and all major surveys of particular objects, e.g. emission-line stars, planetary nebulae, galaxies and galaxy clusters, continue to be based on Schmidt plates. Some discoveries also include the archival use of Schmidt plates: a recent example is the identification of the progenitor of the bright supernova 1987A in the LMC with a surprisingly blue colour and early-type spectrum; this "observation" had a direct impact on model calculations of late

evolutionary stages of heavy stars. Astrometry. Due to the absence of an all-sky, accurate astrometric net with sufficient density for CCD work ( $\sim 1$  star per square arcmin) -the HST Guide Star Catalogue may be well suited for guiding purposes, but not for astrometry at the sub-arcsecond level -the only existing method for the determination of accurate astrometric positions of objects too faint to be observed with meridian circles is to use Schmidt plates on which a sufficient number of bright astrometric standards are visible. The absolute position of a faint object (e.g. in the FK5 system) must either be measured directly on a Schmidt plate or, if it is only visible in a narrow-field, deep CCD frame, by transfer of local secondary standards from a Schmidt plate to the CCD field. Photometry. Absolute photometric accuracies of the order of  $\pm 0.07$  mag can normally be achieved on well-exposed photographic plates for point-like objects which are at least 3 mag above the plate limit. It is possible to improve this value somewhat by having many photometric standards in the immediate area, but the intrinsic non-linearity of photography and the need to establish a unique intensity calibration for each plate restrict the ultimate level of precision. This compares to  $\pm 0.01$ - $0.02$  mag in good CCD frames (total counts per object  $> 10,000$ ) and illustrates an important limitation of the photographic method. However, the possibility to measure on the same plate a very large number of objects makes Schmidt photometry ideal for statistical investigations which do not demand the highest accuracy. Variability. Photographic Schmidt exposures are a rich source of discovery of variable objects and a base for many long-term variability studies. They include photometric variability of quasars over many decades, the large-scale investigation of variable stars in selected Milky Way fields and, in particular, the identification of faint, high proper-motion objects, e.g. white dwarfs. So far the variability has most often been established by the use of blink comparators, but more recently automatic methods (computer-"blinking") have begun to produce important results. Large Structures. The identification and study of large structures, like Milky Way cirrus and clusters of galaxies at intermediate distances, is obviously facilitated by the availability of wide-field exposures. The importance of all-sky work is amply illustrated by the scientific impact of satellite observatories; it has a direct bearing on the study of the largest structures

in the Universe and its evolution. While the CCD technique lets us study the finest details in individual objects, the establishment of large samples with clean definition criteria is only possible with wide-field instruments. When combined with the high angular resolution and comparatively faint limiting magnitude of present, large Schmidt telescopes, it is also possible to observe at the same time smaller details within larger structures; some examples are the overall structure of the Gum Nebula, the intricate network of thread-like nebulae in the area of Carinae, and the complex of interstellar reflection nebulae, near the south celestial pole. In this connection must also be mentioned the objective prism technique, which makes it possible to obtain large numbers of spectra of stars and small nebulae simultaneously and which is unsurpassed for deep studies of the overall stellar distribution in the Galaxy. In summary, it is obvious that photographic wide-field observations with Schmidt telescopes play an important role in current front-line observational astronomy and would also continue to do so in the near future.

### Limitations of the Methods

It is now of interest to look more closely at the critical limitations which are inherent in the two types of detectors in order to better judge how the introduction of a CCD into one of the existing large Schmidt telescopes may influence its observational potential. It is important to emphasize that here it is not the question of installing a small CCD chip into a large Schmidt telescope; this would immediately rob it of its unique wide-field capability, the importance of which has been demonstrated above. Nor is it likely that it would be technically feasible to have a dual system, with the possibility to change between large plates and the CCD camera and its cooling system. The mechanical adjustment of a large Schmidt telescope is extremely critical and frequent exchanges would most certainly lead to inferior performances. Present Schmidt telescopes are obviously optimized for photography, but it would of course be interesting to construct smaller, intermediate-field instruments which are optimized for the largest available CCDs.

### Photography

Regular users of Schmidt telescopes are well aware that in wide-field work, the quality and the astronomical value of the photographic plates depend on a very large number of factors, not all of which can be fully controlled. In the case of sky surveys, of which those in the northern (Palomar) and the southern sky constitute the most comprehensive observing

programmes presently undertaken with photographic Schmidt telescopes, the quality of the final product, i.e. the Atlas copies, is among others influenced by the seeing, the mechanical performance of the telescope, the processing and copying of the plates and the archival conditions. Lack of technological progress. Perhaps the most serious problem in present-day astronomical photography is that for quite a few years, no new, improved emulsions have become available; the latest major advance in the photographic field was the introduction of the emulsion in 1970, followed some years later by another emulsion. Likewise, most of the photographic techniques which are now in use were already fully developed more than 10 years ago. Thus the field of astronomical photography, in the technical sense, is in a period of stagnation, with correspondingly little incentive to new investments in photographic techniques by the observatories. This is also reflected in the decision not to include any photographic equipment in the new generation of super telescopes, for instance in the ESO 16-m Very Large Telescope, even though the large focal fields could only be fully covered by a photographic plate. The use of multiple-slit techniques, e.g. in the planned fibre optics spectrographs, will only permit the simultaneous registration of a minuscule fraction of the light that is collected and focused by the telescope. Extraction of information. Next, the extraction of information from photographic plates has always been a bottleneck in Schmidt work. However, there are positive developments. During the past years, it has been increasingly realized that photographic plates contain a greater wealth of information than thought before, and that their potential has not been exhaustively exploited in the past. This is demonstrated by the impressive results obtained with the photographic amplification, masking and stacking techniques which now make it possible to recognize and measure extremely faint objects which cannot be perceived by direct, visual inspection or even by microphotometric measurement of the plate. At the same time, much effort has been put into improving the extraction of data from photographic plates by means of fast scanning microphotometers, but there are only a few devices in the world in the technological front-line which are also in regular use. Main areas of future improvement are increased speed and extension of the dynamical range by means

of better sources of illumination and faster A/D converters. Archival storage of photographic plates. Photographic plates are stored under very different conditions at the world's observatories. While some plate vaults have advanced climatic control, in other places plate libraries form part of book libraries and the plates are therefore subject to variable temperature and humidity. Old envelopes are often dangerous to plates, since they contain traces of bleaching and/or colouring agents. It is unfortunately also true that not all plate archives profit from an associated efficient retrieval system and at some observatories there is no access to the plates for outside researchers. In this connection, there are interesting developments which aim at the digitization of existing photographic sky surveys, although it is not yet technically possible to store the complete information of a major sky survey in digital form at an affordable cost. However, the availability of computer-readable digital copies with less resolution and smaller dynamical range than the original survey plates will be very useful for many purposes, e.g. the preparation of finding charts for observations.

CCDs When compared to photography, the main drawbacks of the CCD technique are the smaller frame size and the archival problems, while the total cost may play a smaller role. Field size. The currently largest CCDs in common use in astronomy have  $2000 \times 2000$  pix fields.  $4000 \times 4000$  pix arrays have been experimentally tested but it will still be some time before a reasonable number become available. It seems realistic that mosaics of up to four  $2000 \times 2000$  pix CCDs may be available to observers in five years time. Such a device would cover about  $1/25$  of the Schmidt field. With the present sensitivity ratio, the total CCD exposure time necessary to cover the same field to the same limiting magnitude as one photographic plate would be similar; this however, does not take into account the non-negligible read-out time of the enormous data quantities. The curved focal field in present Schmidt telescopes may be another obstacle. In the ESO  $f/3$  Schmidt, the focal position is critical to within  $-10$  mm, compared to a curvature of  $-25$  mm from the centre to the corner of a  $3 \times 3$  cm,  $2000 \times 2000$  pix CCD chip; this may necessitate the insertion of a field flattening lens. The proper recombination of the individual frames into a larger unity, when needed, may also constitute an added difficulty.

Archiving. Probably the most serious problem is that there is still no cheap,

universally available storage medium for CCD data. Magnetic tapes are still used in most places to provide a temporary record of raw CCD data and the rapid accumulation of these tapes, in particular because most CCD exposures are comparatively short, is of major concern in many places, including the European Southern Observatory. Unless the tapes are transcribed to a more stable medium, e.g. optical disks, it must be feared that a significant fraction of the valuable data will become unreadable and therefore lost within a few decades. Also the long-term stability of optical disks is still relatively unknown. The problem of organizing and retrieving information in a data base of several Tbytes should not be underestimated either.

**Cost.** A complete, top-quality CCD system for use in astronomy implies a capital expense that may exceed USD 100,000, and therefore be beyond the financial reach of most smaller observatories. A CCD can of course be used to obtain an unlimited number of frames, and when the total costs are calculated, the need for archiving of the CCD data must also be taken into account. This involves the time-consuming transfer to the final storage medium and the cumulative cost of magnetic tapes and/or optical disks as well as the associated computer hardware and software. Although in photography the capital investment at the telescope and in the plate processing laboratory may be smaller than the above figure, the total costs are not necessarily so. The 1991 price of one 14 x 14 inch spectroscopic plate is below 100 USD, but in recent years a manufacturer has repeatedly increased its prices for astronomical emulsions. For a qualified astronomical exploitation of the plates, it is furthermore necessary to acquire (or have access to) a fast measuring machine and a photographic laboratory equipped for advanced plate manipulations.

**Conclusions** In summary, it appears that the photographic technique has virtues which make it better suited than CCDs for wide-field, high-angular resolution Schmidt telescope work, at least until further notice. Notwithstanding the shortcomings of the photographic plate, its large size and archival properties make it indispensable for the permanent recording of large sky areas during optical sky surveys and patrols. As mentioned above, there is a great current interest in all-sky work and the intimate collaboration between such instruments working in different wavebands, on the ground and from orbiting satellites, is absolutely necessary for the study of the largest structures in the

Universe and therefore of its evolution. For more than 100 years, the photographic technique has guaranteed a continuous record of the visible sky, of immense and lasting value for astronomical research, now and in the future. Sky surveys and patrols must continue, lest future generations of astronomers will blame us for not having done our historical duty. When carefully processed and properly stored, photographic plates can last for centuries, while there is at the present no guarantee that even a fraction of the many CCD frames obtained during the past ten years will also be accessible, say, 100 years from now. It must be stressed that the present conclusion does not apply to work at lower angular resolution and with smaller instruments; there are several examples of the great utility of wide-angle CCD work, for instance the extensive monitoring of Comet Halley's tail with a 640 x 1024 pix camera at La Silla in 1986. It would be highly desirable in the future to complement large Schmidt surveys with very-wide field patrol exposures by specialized low-resolution CCD cameras. There is little doubt that CCDs and other digital detectors will ultimately replace the photographic plates in all instruments, and also in the large Schmidt telescopes. But this step should only be taken when these detectors have become big enough not to compromise the efficient and exhaustive exploitation of the unique scientific capabilities of wide-field Schmidt work and when the CCD archiving problem has been satisfactorily resolved.

Optical Gyros for Astronomical Telescopes

Pointing and tracking of telescopes requires high-precision angular encoders. Up to now classical rotary encoders have been used for this application. Their implementation in modern large telescopes with alt/azimuth mounts becomes rather difficult, because they have to be installed on the rotation axis, which has to be clear and free of any obstruction for the Nasmyth or coude light path. In the NTT the largest monolithic encoders ever built are used, with a clear inner diameter of 50 cm. Its glass ring, which contains the division scales, has an outer diameter of 70 cm. For the VLT, with Nasmyth beams of approximately 1 m diameter, this type of encoder is not realistic any more and alternatively engraved steel scales mounted on precisely aligned cylinders will be used. For these reasons, optical gyros have been proposed at ESO as an alternative solution, already in 1986. They have not to be mounted on the rotation axis because they would control the telescope (pointing and tracking) like an

inertial navigation system used in airplanes, satellites, submarines, etc. In order to assess this new approach and to demonstrate its usefulness for astronomical telescopes, ESO launched a feasibility study in April 1990. Optical gyros are based on the Sagnac effect. Two counterpropagating optical waves are travelling inside a ring interferometer and then combined and brought to interference. If the interferometer is not in rotation then the travel time for the two optical paths are the same and the interference pattern at the output is stationary. If the interferometer is rotating around an axis perpendicular to its plane, then one light wave sees its path shortened and the other one sees it elongated. This classical view leads to correct results for the path differences but a detailed physical description has to be based on the general theory of relativity. A theoretical investigation of this concept in view of the specific requirements in astronomical telescopes and came up with a proposal for a fibre/laser gyro combination in order to cover the pointing as well as tracking phase in telescope operation. A first prototype for the fibre gyro part has recently been completed which would already meet the VLT requirements for tracking with an accuracy of better than 0.1 arcsec over 30 seconds, and a resolution higher than 0.02 arcsec. Commercially available systems currently do not provide this accuracy and resolution. The construction for an improved version which will be delivered to ESO and tested on the NTT next year has just started. This fibre gyro will be combined with a commercially available laser gyro covering the faster slewing and pointing operations of the telescope. A Litton LTN-90 laser gyro system has been considered for this. Multi-Object Spectroscopy with an Automatic Fibre Positioning System in a One-Degree Field First Technical Run of MEFOS at the Prime Focus of the 3.6-m Telescope Introduction Several scientific programmes require the acquisition of a large number of spectra to build up a statistically significant sample of data. Typical examples of this category are studies of galaxies in clusters or in the field and surveys of QSO candidates and peculiar stars in selected galactic regions or in nearby galaxies. In the last decade, optical fibres have been successfully used in gathering the light of different targets spread over the field of a telescope to a common spectrograph slit and thus to speed up the process of data collection. In the first generation of instruments of this type, the fibres -typically between 40 and

100 -are manually inserted in predrilled plates mounted at the focal plane of the telescopes. The ESO facility OPTOPUS is based on this principle and is successfully in operation at the La Silla 3.6-m telescope since 1986 as a common user instrument. Its performance has recently been upgraded with the introduction of two new fibre bundles and of a new F/6 collimator. The plate drilling operation has recently been transferred to the ESO workshop at La Silla. In the second generation of fibre instruments, which came into use more recently, the positioning of the fibres in the field is done automatically at the telescope in order to skip the need of the predrilled plates and to retain real-time control of the fibre position. Systems of this type have been prototyped by Hill and Lesser at the Steward Observatory, by Parry and Gray at the Anglo-Australian Telescope and by Ingerson et al. at Cerro Tololo. In 1989 ESO concluded an agreement with the Observatoire de Paris to have an automatic fibre positioning device built for the prime focus of the 3.6-m telescope. The concept of the instrument was inspired by the systems built at the Steward Observatory and at Cerro Tololo, but it includes some original features. It has the advantage of shorter setting time -the arms can move in parallel -and it gives the possibility to easily correct for atmospheric diffraction and to improve the sky subtraction by switching between object and sky fibres during the exposures. The agreement foresees that the group led by Paul Felenbok at the DAEC department in Meudon will design and build the fibre positioning and target acquisition/guiding units while ESO will deliver the fibre optics bundle and take care of the interface to the telescope and the existing grating spectrograph. The instrument schedule foresees the final installation at the telescope by the last quarter of 1992. In January 1991, a prototype version of the instrument named MEFOS (Meudon-ESO Fibre Optics Spectrograph), was tested at La Silla. This article gives an overall view of the instrument and briefly reports on the results of the first telescope test.

**The MEFOS Project** At the ESO 3.6-m telescope prime focus, using the triplet corrector, a flat, corrected field of one degree diameter is available for faint-object spectroscopy. MEFOS is designed to pick up targets over this field. Four main subsystems can be identified: the positioning arms, the fibre optics, the acquisition and guiding system and the spectrograph. There are 30 positioning arms arranged in a circle at the edge of the field, each of them carrying two

spectroscopic fibres and one image fibre bundle. The F/3 input aperture of the beam at the prime focus is well suited for the best performance of the fibres as far as the focal ratio degradation is concerned. No front lenses are needed at the fibre top ends. The bundle of spectroscopic fibres is guided to the Cassegrain cage where it is interfaced to a modified Boller and Chivens spectrograph. This could be substituted in the future by a dedicated bench spectrograph mounted in a stationary configuration in one of the coude rooms of the telescope. The output ends of the image fibre bundles are projected onto a CCD viewing camera fixed to the structure of MEFOS and used for target recognition and centring.

### Positioning Arms Optical Fibres

The two single fibres used for spectroscopy have the same projected aperture on the sky: 2.6 arcsec (135  $\mu$ m). The fibre length needed to link MEFOS to the spectrograph located in the Cassegrain cage is 21 m. For the first run at La Silla we used Polymicro FBP fibres. This type of fibre shows a flat transmission between 500 and 1200 nm. To increase the efficiency in the blue wavelength region (350-450 nm) we will use in the future the so-called "wet" fibres. As shown by the dashed line in Figure 4, this kind of fibre is much more transparent in the blue but exhibits water absorption bands in the near-infrared region. A short CCD integration then gives subimages of all 30 targets. These are processed to compute the exact object positions with respect to the single spectroscopic fibres on the corresponding arms. Finally the arms are moved by the computed offsets and the integration of the spectra can be started. The acquisition camera uses a Thomson CCD with 1024x1024 pixels of 19  $\mu$ m, cooled with a two-stage Peltier device to about -60°C. Lower temperatures are not needed because the exposure time will never exceed a few minutes and the read-out noise is small compared to the sky photon noise. A 12-bit A/D converter is used in the commercial CCD control camera leading to a sufficient dynamical range and to a fast read-out. As shown in Figure 1, the acquisition system is placed in such a way that a separate TV camera can monitor the centre of the field by means of a third photo-objective. A movable 45° mirror may be inserted to project the image fibres on this TV camera. In this position, the automatic guiding of the telescope can be achieved with one image fibre pointing to a conveniently bright star in the field. If such a star cannot be seen on any of the image fibre bundles, one of the arms must be moved to find it. Control

of the Instrument An Olivetti M300 PC/AT is used for the general control of the positioners and the Acquisition and Guiding module. The arms are driven by a central master card located in the PC. This card is connected through two optical fibers to slave cards located on each positioner. Both cards, master and slaves, are based on 8 bit 80451 and 80535 microprocessors, respectively. The coordinates of the arm positions are stored in the master card. The latter questions and sends commands to all the arms; once well received and acknowledged, these commands are executed by the arms. The arms move therefore simultaneously and the maximum time to set up a field configuration from the "parking" position is less than 5 minutes with a repeatability of 10 mm (0.19 arcsec). The master microcontroller continuously scans the whole system. By questioning the master card, the user can observe in "real time" the dynamic state of the system. Communications between the master and the slaves are done in serial at a data rate of 375 Kbaud. The two optical fibres (transmit and receive) have a length of 75 m and are provided with an error-detection procedure. Spectrograph The Boller and Chivens spectrograph has been adapted to the use with MEFOS. The standard F/8 parabolic mirror used as collimator and the Schmidt camera have been replaced by two fully dioptric elements, an F/3 collimator and an F/2 camera, respectively. With this new configuration the fibre-spectrograph matching is optimal. As was mentioned above, the light losses at the central obscuration of the camera are avoided and the focal ratio degradation at F/3 is minimal. During the first test observation in January 1991, the spectrograph detector was a Tektronix CCD with 512x512 pixels of 27 mm. The fibre output ends project a diameter of 86 mm or about 3 pixels on the detector. The usual complement of gratings and corresponding dispersions of the B & C spectrograph in the OPTOPUS configuration is available to the observers. First Technical Run at the 3.6-m Telescope The first test run of MEFOS at La Silla took place from January 30 to February 7, 1991. The final instrument structure was used but with nine positioning arms only. The goal of the observing run was to check the telescope interfaces and to practice all instrument mounting and adjustment procedures. The fibre output slit could be mounted either at the spectrograph collimator or at a photomultiplier to verify the accuracy of the fibre centring. The whole system

(including the read-out of the photomultiplier) was controlled by the instrument PC in the Control Room. The spectral calibrations and flat-fields were performed using the Optopus flange mounted, as usual, on the Cassegrain adapter. The lamps sent the beams directly to the prime focus through the central hole of the main mirror. They were controlled together with the spectrograph CCD by the OPTOPUS software package running on the HP 1000 telescope computer. Apart from minor difficulties in the mechanical installation and in the control software, the main problems were encountered in the object acquisition: at the beginning the arms were not able to reach the correct positions. This was traced down to a slightly erroneous value of the scale we were using. After this correction, the final position of the arms was still not fully satisfactory because the programme did not yet include the field distortion. Nevertheless, once the objects were brought inside the image fibres and analysed with the acquisition programme, the arms could send the spectral fibres to the objects with relatively good accuracy: better than 0.4 arcsec. Two factors contributed to this uncertainty: the spherical aberration produced by the non-perfect alignment of the triplet corrector, and small drifts in the tracking of the telescope during the acquisition exposure time. In the laboratory the procedure of target acquisition and displacement of the arms to put the spectral fibre in front of the object yields an accuracy of better than 10  $\mu$ m or 0.17". In the last three nights of the run, a number of scientific exposures were obtained. The most important exposure was on a field of galaxies with magnitudes between 17.5 and 18.6. This field had been observed before with OPTOPUS. The field acquisition exposure time of 5 min and the spectral exposure time of 1 hour proved to be sufficient for the purpose. The spectrum displayed in Figure 5 is from a galaxy of  $m_a = 18.6$ ,  $z = 0.06$  and reaches a signal-to-noise ratio of 50. An actual measurement of the relative efficiency of MEFOS and OPTOPUS is almost impossible because of the strong dependence on seeing. A computation which takes into account telescope, fibre and spectrograph effects indicates that MEFOS should be approximately 25 % more efficient than OPTOPUS. News on ESO Instrumentation EMMI EMMI, the ESO Multi-Mode Instrument, is in regular operation at the Nasmyth B focus of the NTT since November 1990. In March and April of 1991 part of the EMMI team was again on the mountain for a number of upgrades on the instrument. These

are shortly summarized below.

### Multi-Object Spectroscopy

The operation and the first results of the MOS mode of EMMI have been described in the Messenger. Further work was needed to refine the object selection software, for slight modifications of the hardware and to prepare a user interface. The work is now completed and the mode is in operation.

### Medium-Dispersion Spectroscopy with the Dichroic

The DIMD mode is now also in operation. In this configuration the slit is fed by a wide-band mirror instead of the blue- or red-optimized mirrors and the blue and red beamsplitter prism below the slit is replaced by a dichroic prism. All types of coatings represent state-of-the-art coating technology. The absolute efficiencies as measured in the ESO optical laboratory are shown in the figure. The EMMI control software fully supports the DIMD mode and allows parallel exposures of the two CCDs.

### A Ford 2048 CCD Installed in the Red Arm

In the first six months of observations, the red arm of EMMI operated with a coated 1024 TH CCD as a detector. A number of complaints were received because of the compressed scale, of the field limitation and of the heavy fringing at red wavelengths due to the coating. In April a 2048 Ford CCD was installed: this now gives the full field of the instrument with a better scale. The cosmetic is excellent and the CCD is uncoated. The efficiency is lower than for the coated Thomson CCD below 450 nm only, a region where the red arm is not really designed to operate. Still there are applications where the lower readout noise of the Thomson instead of the Ford is an asset. While the discussion between the two detector options goes on, we plan to install a new CCD next year.

### Upgraded Version of the User Interface

EMMI is a complex instrument which offers on-line two large-size CCD detectors and very different modes of observations, from echelle spectroscopy to multi-object spectroscopy. Given this background it is not surprising that the occasional user needs time to learn how to deal in the most efficient way with the instruments and its control software. It is the opinion of the author that instruments like EMMI should be ideally operated in a service mode to exploit in the most efficient way their capabilities, but it is very difficult, besides being expensive, to find and train the right operators for this type of work. In the meantime, we have to rely on the good will and the unlimited resources of the visiting astronomers. The excellent scientific results obtained at the NTT by many visitors prove that the learning process can be

very fast and effective. In these first months of operation the control software, initially not bug-free, has been consolidated and a new, more friendly version of the user interface was installed in April. In November 1991, a further upgraded version will be installed, again with the goal to speed up some operations and to make the system simpler to use. Thanks to the work of the software group, a workstation has also been added to the system for data acquisition and to operate MIDAS at the telescope. This is of considerable help because the computer-based data-analysis system cannot handle large-size CCD images. Planned Improvements of EMMI Two detector changes are foreseen to further improve the performance of the instrument. In November of this year it is planned to install a thinned 1024 TK CCD in the blue arm to increase the efficiency and to increase the field. In the middle of 1992 it is planned to install a 2048 TK device in the red arm to increase the efficiency in the 400-500-nm range.

SUSI With the installation of EMMI at the Nasmyth B focus of the NTT in November 1990, the possibilities of high angular resolution imaging at the NTT had become limited. With the installed optical cameras and CCD detectors, the EMMI pixel size is 0.29 arcsec in the blue arm with the 1024, 19-mim pixel Thomson CCD and 0.35 arcsec in the red arm with the 2048, 15-mim pixel Ford CCD. With these scales the CCDs cover the total corrected field of the instrument. It is an optimal compromise between the requirements of the different modes of observations of the instrument, but the stellar images are not sampled in an optimal way when the seeing is better than 0.7 arcsec approximately. The experience of the first year of operation of the NTT has shown that this happens for a relevant fraction of the observing time, and therefore unique opportunities for imaging at high angular resolution could be lost. To fill the gap, ESO developed in a 6-month crash programme a dedicated imaging facility named SUSI, the Superb Seeing Imager to be mounted on the adapter at the Nasmyth A focus and to be used as a standby mode of EMMI for observations in unique seeing conditions. In April 1991, SUSI was installed at the NTT and immediately offered to the visiting astronomers whose programmes call for high angular resolution. The user interface will be finalized in November 1991. The change from EMMI to SUSI can take as little as 10 minutes. The instrument at the telescope has a very simple and compact structure. At the centre of the field, a three-position

mirror unit takes care of directing the light to an optical CCD or to a "visiting" detector or to let it finally pass through to IRSPEC. The CCD dedicated mirror used at present is aluminized. In November, a wide-band, high-efficiency coated mirror will be installed. A filter wheel with 8 positions is placed in front of the CCD shutter in the converging beam of the NTT. The internal diameter of the single filter cell to be fitted in the wheel is 60 mm. The CCD detector presently used is a thinned TK 1024, 24-mim CCD. This leads to a pixel size of 0.13 arcsec and a field of 2.2x2.2 arcminutes. The quantum-efficiency curve as measured is given in the figure. The design of the mirror unit and of several other parts profited from the experience on the EMMI functions. Still, to have the instrument designed, built by external firms, integrated and tested in Garching and later in Chile within little more than 6 months was a major achievement for which the SUSI project team is to be congratulated. That the effort paid off can be seen from the two examples of astronomical observations.

#### Aluminium Mirror Technology at ESQ: Positive Results Obtained with 1.8-m Test Mirrors

The aluminium alternative for the manufacturing of astronomical mirror blanks developed initially around the NTT project. A series of tests on 500-mm samples led to the conclusion that a 4-metre-class aluminium mirror was feasible, with a similar level of quality as a glass mirror. This option was seriously considered for the NTT active primary mirror but finally canceled because of schedule problems. In addition to being much less expensive than glass, aluminium also presents advantages such as reduced fragility, easy machining, possible repair and excellent thermal conductivity. The latter almost eliminates the risk of thermal gradients (in normal operation) and would improve the efficiency of thermal control (mirror seeing). The main problem is that few data on long-term stability exist. Since 1968, Merate Observatory runs a 1.4-m telescope with an aluminium primary. Measurements carried out by ESO in 1982 showed a total aberration of less than 1 fringe, well within the spatial frequency range of an active support. Development continued within the framework of the VLT programme. Since the technology selected for the NTT (casting) does not seem extrapolable above four metres, different manufacturing processes had to be investigated. Initial tests on 500-mm samples led to the selection of two promising techniques: build-up (BU) welding

and electron-beam (EB) welding. It was consequently decided to pursue investigations on intermediate-sized blanks of 1.8 m diameter. Two blanks were purchased, one from Linde (Germany, build-up welding) and one from Telsa (France, electron-beam welding). The principles of both technologies are shown in Figures 1 and 2. Build-up welding consists in building up the complete aluminium piece by continuous deposition of a welding seam. After selection of the proper alloy and optimization of the welding parameters, excellent homogeneity of the crystalline structure and extremely low porosity could be achieved. Although this technique is well known with steel and applied to pieces in the 10-m dimension range (pressure vessels), it was surprisingly less known by most aluminium manufacturers. A picture of a 500-mm build-up aluminium mirror blank is shown in Figure 3. Electron-beam welding is a well-mastered welding technique which consists of fusing pre-assembled pieces by means of an electron gun of sufficient power. The technique is well mastered in industry and the demand for scaling the technology up to large diameters already exists. The pieces to be assembled can be cast, forged or rolled. An appreciable advantage of EB welding is that no extra material is introduced which reduces the risk of inhomogeneity. For the manufacturing of an astronomical mirror the most interesting process is to weld forged segments, since forged aluminium pieces show excellent homogeneity and very low porosity. A picture of the 1.8-metre EB welded mirror blank shortly after welding of a radial joint is shown in Figure 4. All problems encountered at the blank manufacturing stage could be efficiently solved, and no limitation could be found in a possible extrapolation to 8 metres. Independently of the ESO activity, experiments in the 8-metre range are carried out in the framework of the Eureka-funded LAMA project. Two 4-metre-long segments which will represent 2/16 of an 8-metre mirror blank will be EB welded. After manufacturing, the 1.8-metre blanks received thermal treatments which included annealing and cryogenic stabilization. After machining, the blanks were 1.8 metre in diameter, 300 mm thick, with a flat back and a f/1.67 spherical concave surface (within an accuracy already in the micron range, thanks to computer-controlled machining). Both blanks were shipped to Reosc for optical figuring. Rough grinding was followed by nickel coating, subcontracted by Reosc to Tecnol. Nickel coating

consisted in the deposition of a nickel layer by a chemical process. The thickness of the layer is in the range of 0.1 mm. A nickel coating is required because aluminium is not directly polishable, at least within the same level of cosmetic quality as glass or nickel. After nickel coating the mirrors were fine ground and polished spherical, with a radius of curvature of 6 metre ( $f/1.67$ ). Because of the innovative aspect of the experiment, ESO required that emphasis be put on the accuracy of surface measurements instead of the accuracy of the optical surface itself. However, both mirrors were polished well within specifications and according to Reosc, there would be no problem in achieving the same standards of quality as with glass. For what concerned bubbles and inclusions, both aluminium blanks were comparable to glass blanks. After polishing the effect of these bubbles and inclusions on the optical surface was totally negligible. The high expansion coefficient of the substrates did not cause significant problems, neither did the bimetallic effect between the aluminium and the nickel. In polishing nickel coated aluminium mirrors, the danger is not with the adherence of the nickel layer but with the risk of breaking through the nickel coating, which required that the thickness of the coating be monitored. This risk emphasizes the requirements on accurate machining prior to coating, and on the uniformity of the nickel layer. After acceptance of the mirrors, their stability towards ageing was checked. Thermal cycles followed by interferometric tests at centre of curvature were ordered from Rease. One cycle consisted in varying the temperature from ambient temperature down to -20 degrees centigrade, then up to 40 degrees and back at ambient temperature, within about 24 hours. Both mirrors underwent 32 cycles, with intermediate checks after 4, 8, and 16 cycles. The final results are summarized in Figures 5 and 6, which show the surface maps before and after 32 cycles. Surface errors were measured interferometrically at centre of curvature with a sampling of about 8000 points on the mirror (100 points across a diameter). The surface maps presented in Figures 5 and 6 result from the averaging of 50 interferograms each; the standard deviation is about 1 nanometre on the RMS surface error. Even on knife-edge images, no fine structure linked to the welding seams could be detected. The most important figures are the variations of the surface errors rather than their absolute values, and above all the variation of high

order effects, defined as surface errors after the 3rd and 5th order optical aberrations have been mathematically removed. While variations of the surface error function are observed, the global optical quality remains stable. The variation of the surface errors (axisymmetrical for the Telas mirror, astigmatic for the Linde) are less than one fringe and would be fully compensated by a simplified active support. They were detected after the first inter-ESO, the European Southern Observatory, was created in 1962 to establish and operate an astronomical observatory in the southern hemisphere, equipped with powerful instruments, with the aim of furthering and organizing collaboration in astronomy. It is supported by eight countries: Belgium, Denmark, France, Germany, Italy, the Netherlands, Sweden and Switzerland. It operates the La Silla observatory in the Atacama desert, 600 km north of Santiago de Chile, at 2,400 m altitude. where fourteen optical telescopes with diameters up to 3.6 m and a 15-m sub-millimetre radio telescope (SEST) are now in operation. The 3.5-m New Technology Telescope (NTT) became operational in 1990, and a giant telescope (VLT=Very Large Telescope), consisting of four 8-m telescopes (equivalent aperture  $\sim 16$  m) is under construction. Eight hundred scientists make proposals each year for the use of the telescopes at La Silla. The ESO Headquarters are located in Garching, near Munich, Germany. It is the scientific-technical and administrative centre of ESO where technical development programmes are carried out to provide the La Silla observatory with the most advanced instruments. There are also extensive facilities which enable the scientists to analyze their data. In Europe ESO employs about 150 international staff members, fellows and associates; at La Silla about 40 and, in addition, 150 local staff members. The ESO MESSENGER is published four times a year: normally in March, June, September and December. ESO also publishes Conference Proceedings, Preprints, Technical Notes and other material connected to its activities. Press Releases inform the media about particular events. The conclusion is that in visible light ( $\lambda=500$  nm) the stability of both mirrors towards ageing cycles is of the order of  $1/25$  for the overall rms surface errors and  $1/70$  for the high spatial frequency rms surface errors, a very positive result. Moreover, the changes occur during the first cycles and the surfaces remain almost perfectly stable afterwards. A question still open is the

homogeneity of the coefficient of thermal expansion (CTE). Although the effect of possible inhomogeneities in the range of 1 to 5 % of the nominal CTE (a very generous tolerance) could probably be compensated with active optics (unless they are very localized), a definite answer should preferably result from a series of tests at different temperatures. This experiment has brought great confidence in the aluminium mirror technology. Both technologies, BU and EB welding, proved adequate for the manufacturing of 2-metre-class astronomical mirrors. Not only were the optical tests very successful, even more important is the fact that extrapolation to larger diameters now seems possible.

First VLT Call for Proposals Introduction On 1 August 1998, ESO will release the Period 63 call for proposals for all ESO telescopes. For the first time this will include the 8.2-m Unit Telescope 1 of the VLT. The deadline for proposals will be 1 October 1998 for all telescopes which represents an increase of one month in the preparation period from previous rounds. The OPC will meet in the first week of December 1998 for both UT1 and La Silla telescopes. Period 63 operations will commence on 1 April 1999. Proposals for Period 64 will be called for on 1 February 1999 and close 1 April 1999. During Period 63, UT1 will offer instruments in both Service Mode and Visitor Mode. It is ESO's intent to ultimately offer Service Mode observations for all most frequently used modes of instruments on the VLT in order to permit the best utilisation of the unique properties of the Paranal site and to guarantee a minimum level of consistency to the archived data. As we accumulate experience on the most effective use of the instruments and develop adequate software tools, we plan to support service as well as visitor observing with automated calibration pipelines to initially remove instrument signatures and ultimately to provide physical quantities. Initially, such pipelines will exist only for the simplest modes of use of the instruments offered. Service observing can be carried out without the benefit of such standardisation in the same manner that visitor observing is carried out. This mode of operation is much more manpower intensive and will be offered on a best-effort basis. ESO will work closely with the user community to develop service and visitor mode operations and resources that will maximise the scientific return of the VLT.

In Service Mode, successful principal investigators will specify observation programmes as a series of Observation Blocks (OBs) using Phase 2 Proposal

Preparation (P2PP) software tools. These OBs will be executed by ESO staff astronomers on UT1 to a schedule dictated by OPC ranking and prevailing conditions. In Visitor Mode, astronomers will again construct OBs but will journey to Paranal and be present when they are executed. Instruments for UT1 in Period 63 ESO plans to offer FORS1 and ISAAC on UT1 beginning in April 1999. Detailed descriptions of these instruments can be accessed from the ESO Web page. All VLT instruments are commissioned in two phases. In the first phase, the instrument is mounted on the telescope for the first time and functional tests are made of all operational modes. This phase is followed by an assessment period where instrument performance is evaluated and observing templates are optimised. The second and final commissioning phase sees operational tests of all observing modes to be offered to the community. Each instrument then enters a Science Verification period in which test science programmes are executed under actual operations conditions to assess science performance and readiness for operations. At this time ESO is planning to carry out the commissioning schedule for FORS1 and ISAAC outlined in Table 1. The final set of instrument modes and the measured instrument performance on UT1 offered to the ESO community on 1 April 1999 will depend on the results of the commissioning and science verification processes. Principal investigators may have to make modifications to observing programmes in early March based on the outcome of commissioning. Tables 2 and 3 present the FORS1 and ISAAC instrument modes ESO plans to commission by 1 March 1999 and offer during Period 63. Application Process for Period 63 Observing proposals for all ESO facilities on La Silla and Paranal will use a revised application form. Although revisions were necessary to provide VLT support, ESO took this opportunity to streamline the form in many areas and to provide the capability to include Postscript figures. As usual, the La Silla Call for Proposals will only be available from the ESO Web site. A separate VLT Call for Proposals will be published both on paper and on the Web. The VLT Call for Proposals will contain all the information necessary to complete and submit a VLT observing proposal, including a description of the detailed supplementary documentation available to proposers. Such detailed documentation includes the VLT White Book and the individual instrument handbooks. The VLT Call for Proposals will also

provide the information necessary for proposers to decide whether Visitor Mode or Service Mode is more appropriate for their project. Information about the FORS1 and ISAAC Exposure Time Calculators and how they can be used to support the proposal process will also be provided. Preliminary versions of the FORS1 and ISAAC ETCs are available via the ESO Web site. Astronomers preparing VLT observing proposals should be aware that if they are awarded VLT time, they will be asked to complete a Phase 2 proposal process. During the Phase 2 process, proposers will create detailed descriptions of their observing programme in the form of Observation Blocks (OBs). Visitor Mode PIs will be strongly encouraged to construct their OBs before traveling to Paranal but will not be required to finalise their OBs until after they have arrived there. Service Mode PIs, however, will be required to submit their OBs to ESO before their programme will be executed. More information about this process will be provided in the VLT Call for Proposals. Astronomers planning to propose for VLT time may also wish to read. A detailed guide to preparing OBs using Phase 2 tools will be published in the December 1998 Messenger. Astronomers with further questions regarding the call for proposals for Period 63 are asked to contact the User Support Group at ESO.

In this evening view, obtained about half an hour after sunset, the UT1 telescope structure is seen through the dome slit. The assembly of UT2 is well on its way in the next dome, and the first parts of UT3 are in place in the third. AMBER, the Near-Infrared / Red VLTI Focal Instrument Abstract

The near-infrared /red focal instrument of the VLTI, called AMBER, will operate between 1 and 2.5  $\mu\text{m}$  in a first phase with two UTs. This instrument has been designed for three beams to be able to perform images through phase closure techniques. The wavelength coverage will be extended in a second phase down to 0.6  $\mu\text{m}$  at the time the ATs become operational. The magnitude limit of AMBER is expected to reach  $K = 20$  when a bright reference star is available and  $K = 14$  otherwise. The main scientific objectives are the investigation at very high angular resolution of disks and jets around young stellar objects and AGN dust tori with a spectral resolution up to 10,000.

### 1. Introduction

The interferometric mode of the VLT (VLTI) has always been present in the VLT project. In the last Messenger issue, the ESO Director General has presented the role of ESO in European Astronomy by stressing the important and unique scientific

contribution expected from the VLTI. The VLTI implementation plan has been reviewed in 1995-1996 by ISAC, the Interferometry Science Advisory Committee, who gave its recommendation to the ESO community in The Messenger No. 83. The committee recommended early operations of the VLTI as well as a phased development that would focus in the infrared wavelength range. A new plan has then been proposed by the ESO VLTI team with an updated timetable: operations with two 8-m unit telescopes (UTs) before the end of 2000, with two 1.8-m auxiliary telescopes (ATs) before the end of 2002 and the full complement of 4 UTs and 3 ATs starting in 2003. In 1997, three instruments were proposed: AMBER, a near infrared / red instrument. At this time, AMBER included adaptive optics. MIDI, a thermal infrared instrument. PRIMA, an instrument based on the simultaneous operation of two fields. On 14 April 1998, the VLTI Steering Committee recommended that ESO take the lead to deliver a dual field and stabilized beams, in order to boost the performance of AMBER and MIDI, as early as in 2001, much earlier than expected in the first version of the instrumentation plan. AMBER intends to combine the main advantages of the interferometric instruments for which Europe has acquired experience: the FLUOR instrument and the GIST interferometer. This paper presents a preliminary report on AMBER, where we detail the science drivers, the concept, the expected performance and the overall project organization. The work presented here is the result of two preliminary working groups in addition to the AMBER present group.

## 2. Science Objectives

Of course, a major role in the science operation of AMBER will be played by the limiting magnitude that the system will permit. With such sensitivities, there is a wealth of scientific issues that AMBER will allow us to tackle. Within the project, it has been decided that, at least at an initial phase, the instrument should be dedicated to relatively few topics. An investigation based on criteria of feasibility on one side, and strong interest in the scientific community on the other side, has resulted in a few selected areas which are listed, with a list of typical parameters. It is important to note that AMBER will allow us in principle to cover a wide range of scientific objectives including: the search of hot exoplanets the formation and evolution of stars extragalactic studies which will be our first scientific targets. A detailed description of the scientific rationale cannot be given in full here, but the reader is referred for instance to the

proceedings of the workshops organized by ESO. 3. Preliminary Optical Layout

The incoming beams have their wavefronts corrected by low-mode adaptive optics modules provided by ESO for the UTs and by the AMBER consortium for visible wavelengths. The expected Strehl

TABLE 1: Scientific characteristics for AMBER

The compressed pupil after the cylindrical optics has the same shape whatever the number of telescopes. We plan to use the same spectrograph and detector in all cases. Therefore, increasing the number of telescopes from two to three, and then to four, requires only the addition of one incoming beam with the corresponding optics and the modification of the anamorphic system without changing anything in the already existing beams. The effect of the spatial filter is to reduce the wavefront perturbations to a flux variation in the fibre. If these photometric fluctuations are measured with a good precision, and if the fringe exposure times are short enough, then the fringe visibility can be measured with no experimental bias as has been demonstrated by the FLUOR experiment. The spatial filter and the photometric calibration are therefore necessary to measure visibilities with extremely high accuracy on relatively bright sources. With the spatial filter, the instrument has a field limited by the size of the Airy disk of the individual telescopes.

4. Expected Performances

At this stage of the project, this parameter is still subject to several uncertainties linked to exact specifications of optics throughput, detector and electronics characteristics, as well as fringe tracker and adaptive optics performance. At present, we estimate that AMBER coupled to the VLT UT telescopes should allow us to reach, when a bright star reference is available,  $K = 20$  in broad band and  $K = 15$  at a resolution of 1000. When the interferometer uses the object as a reference, the limiting magnitude is rather  $K = 14$ . At the 1.8-m AT telescopes these limits drop by about 2.7 mag. The upper half of Table 2 gives the limiting magnitudes of AMBER when the instrument detects fringes on the scientific source (self-reference mode). These numbers would also be the limits for the fringe sensor. In this mode, AMBER can detect fringes with  $\lambda/4$  accuracy on stars up to  $K = 14$  with 25-ms exposure time. For stars brighter than  $K = 9.7$ , the fringes can be detected at  $\lambda/40$  accuracy with 4-ms exposure time. In this self-referencing mode, the limiting magnitudes are valid for all spectral resolutions. The dual field allows for off-axis reference stars. The limiting magnitude for these references are

the same as the ones quoted above. The lower half of Table 2 gives the limiting magnitudes on the science object in order to reach a 1% visibility accuracy within 4 hours of observation sliced into 100-second individual exposures. These numbers are given for poorly resolved objects producing maximum fringe contrast in the fringe sensing spectral band. A 0.1 object visibility would lead to a 2.5 mag penalty.

**5. Organisation**

**5.1 Institutes involved** The AMBER consortium is composed of four institutes: Laboratoire d'Astrophysique, Observatoire de Grenoble Observatoire de la Côte d'Azur Osservatorio Astrofisico di Arcetri in Firenze Max-Planck-Institut für Radioastronomie in Bonn. Other institutes provide expert scientists or engineers but do not build or integrate hardware. They are: Institut de Recherche en Communications Optiques et Microondes in Limoges Université de Nice ? Sophia Antipolis Office National d'Études et de Recherches Aérospatiales in Paris Centre de Recherche Astronomique de Lyon.

**5.2 Project structure** The AMBER project includes a principal investigator, a project scientist, a chairman of the science group, a project manager, a system engineer and a co-investigator. AMBER has been divided in a set of working groups, each one in charge of one AMBER subsystem. In addition to the ESO ISAC, there is a science group for each VLTI instrument with the task of identifying and prioritising the key targets in order to maximise the scientific return of the instrument, especially during early operations. In the case of AMBER, the science group includes scientists working on star formation, galaxies, AGN, exoplanets, low-mass stars, AGB stars, Be stars and circumstellar medium. The project scientist is in charge of translating the scientific needs in terms of instrument specifications. He is helped by an interferometry group, for interferometry offers a large range of observing modes and procedures, whose priorities must be analysed and specified by specialists. The subsystem working groups of the AMBER instrument are: Optomechanics Cooled spectrograph, Detector and associated electronics, Instrument control (VLTI interface included), Observations support (observation preparation, data reduction), Testing, integrating equipment and performance tests.

**5.3 Budget** Many elements still have to be defined in the present system definition phase. Therefore, the final budget cannot be completely defined before the Preliminary Design Review in November 1998. The estimated final budget for the

hardware of the infrared part of AMBER for two beams (phase 1) is 4 MF  $\pm 10\%$ . Extension to three beams and to visible wavelengths (phase 2) require additionally 4 MF  $\pm 30\%$  including the adaptive optics modules for two ATs. The preliminary and still approximate general timetable combines the planning for each subsystem and integrates them in a general planning with the following important dates: July 1998: Full definition of the project. Problems have been identified and a concept has been selected. November 1998: Preliminary Design Review. Problems have been solved, detailed system analysis is finished. All interfaces are analysed. Precise timetable is known. April 1999: Final Design Review. All orders can be issued. July 2000: Manufacturing and Integration Review. All subsystems have been integrated and tested and it is possible to start the global integration and tests. December 2000: Shipment to Paranal where, after 3 months of laboratory and siderostat tests on site, we expect to start observations. April 2001: Observations with the UTs.